Many small carriers: the mean-field limit of competitive urban delivery under zone-time charges

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Abstract

Cities face many small delivery operators rather than a few large ones. We study the Cournot-congestion game among carriers whose shares go to zero, on a time-expanded zone graph with multi-stop tours and fleet budgets. The limit is a potential mean-field game whose equilibrium is a Wardrop equilibrium of the zone-hour market. We ask how fast the finite market approaches it and what this means for observation-based charges. Paccagnan et al. (2019) give $O(1/M)$ bounds under strong monotonicity in individual strategies; here profits are linear in routing plans and that condition fails. The aggregate gap is then at most $\frac{1}{2}(\bar{s}\langle m, A\rangle)^{1/2}$, with $\bar{s}$ the largest one-carrier delivery at a zone-hour, and a two-zone family attains this order, so no bound of order $\bar{s}$ with an instance-free constant exists once plans couple zones. For carriers that separate by zone-hour the gap is at most one carrier's delivery (constant one, equality for identical carriers); for replicas it is $O(1/n)$. The observation-only charge converges to the marginal-cost charge, with welfare loss at most $\frac{1}{2}\bar{s}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance. On nine cities with a stylised hourly profile the gap falls like $N^{-0.88}$ between 24 and 1536 carriers, and with few carriers the observation-only charge lowers welfare. With free entry the market-power subsidy attracts entrants and lowers welfare net of fixed costs; in the one-zone linear case the second-best charge is the plain Pigou charge. Through-traffic delay, emissions and noise cost 8.0 times the in-zone congestion charge per delivery.

Keywords: Mean-field games, Wardrop equilibrium, Cournot competition, Congestion pricing, Urban logistics, Convergence rates

1. Introduction

Parcel delivery in large cities is carried out by a mixture of a few large carriers and very many small operators: subcontracted van owners, regional couriers and gig drivers. A companion paper, Dang (2026), under review (the *companion paper* below), studied a city that charges a handful of large carriers playing a Cournot-congestion game in which each carrier solves a vehicle-routing problem. Its leader results (dual implementation pricing and aggregate-response pricing) are driven by market power: with few carriers, the corrective charge contains a subsidy for the deliveries that Cournot carriers withhold.

This paper asks what remains of that picture when the carriers are small. When every carrier's share goes to zero, each becomes a price taker and the market approaches a Wardrop equilibrium of the zone-time market, which is a potential mean-field game on a time-expanded graph (Lasry and Lions, 2007; Huang et al., 2006; Gomes et al., 2010; Guéant, 2015; Calderone and Sastry, 2017). Three questions follow. How fast does the finite market approach the limit, and with what constant? What happens to the city's observation-only charge? What happens to the parts of the theory of the companion paper that depend on the number of carriers,

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namely the belief width that absorbs rounding errors and the vehicle-count regime in which pure equilibria may fail to exist? Section 2 reviews the related literature and states what is new.

Contributions. The paper's main contribution is a precise answer to the rate question for this class of games.

1. A general bound $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \le \frac{1}{2}(\sum_i \|a_i^{\mathrm{W}}\|_m^2)^{1/2} \le \frac{1}{2}(\bar{s}\langle m, A^{\mathrm{W}}\rangle)^{1/2}$, also for pure equilibria with lumpy plans (Theorem 4), and a two-zone family that attains it up to the factor $1/\sqrt{2}$ (Proposition 5). The order $\bar{s}^{1/2}$ is the rate of Paccagnan et al. (2019, Thm. 1(3)), whose Theorem 1(1)–(2) already give $O(1/M)$ under strong monotonicity in individual strategies; that condition fails here because profits are linear in routing plans. Consequently, for carriers whose tours or fleets couple zones, *no* bound of order $\bar{s}$ with an instance-free constant exists; the worst case is of exact order $\bar{s}^{1/2}$.
2. Exact $O(\bar{s})$ bounds under structure, where strong monotonicity is not available: constant one for zone-separable carriers, for convexified and lumpy plans alike, with equality for identical carriers (Theorem 6); constant one in the M-norm for identical carriers under any coupling (Proposition 7); $O(1/n)$ for replicas with an instance-dependent constant (Theorem 8) that is explicit in the smooth case and degrades with the collinearity of the carrier types (Proposition 9).
3. For the city: the observation-only rule contracts at the instance-free rate of the companion paper for both behaviours; its finite-market fixed point equals κ times the Nash aggregate of an auxiliary game, so it converges to the marginal-cost charge at the rate of an aggregate gap (Theorem 12); and its welfare loss is at most $\frac{1}{2}\bar{s}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance (Theorem 13), so efficiency converges at rate $O(\bar{s})$ even where aggregates converge only at rate $O(\bar{s}^{1/2})$.
4. In the limit, Γ-maximin carriers act like a tax $\delta m \odot w$; the observation-only charge then over-corrects, at a welfare cost of at most $\frac{1}{2}\delta^2 \|m \odot w\|^2_{(m+\kappa)^{-1}}$ (Proposition 15); the belief width needed for credal absorption is $O(d\bar{s})$ (Corollary 16). Under vehicle-count congestion the limit has an equilibrium, and it is a weighted potential game exactly when dwell times are rank one (Proposition 18).
5. A computational study on the nine cities of the companion paper, extended to a time-expanded graph with ten hourly periods and a stylised hourly profile, with up to 1536 carriers.
6. Who pays and who enters (Sections 11–12): the welfare split among carriers, buyers and the city; lump-sum fees that fund the market-power subsidy and when carriers accept them (Proposition 21); free entry, with the identity welfare = consumer surplus + city revenue at zero profit and, in the one-zone linear case, the Pigou charge as the second-best charge (Proposition 22); and emissions, noise and through-traffic delay as tour externalities (Proposition 23), computed on the nine cities.

Scope. The limit fits gig couriers, subcontracted van operators and small regional carriers; it does not fit a market dominated by three to six large carriers, for which the atomic model of the companion paper, or a model with major and minor players (Huang, 2010), is the faithful one. We do not treat the major-minor model here. The hourly demand and speed profile is stylised: the repository has no hourly data, and we anchor the profile at published peak-hour congestion values and grade every assumption (Section 9).

2. Related work and what is new

Each subsection ends with what is known, what is new and what we reuse; where a result of ours is a special case or restatement of earlier work, we say so.

2.1. *From Nash to Wardrop: rates and large-market limits*

Wardrop's principle (Wardrop, 1952) and Beckmann's convex potential (Beckmann et al., 1956) describe a population of negligible users. Haurie and Marcotte (1985), as reported by Paccagnan et al. (2019), showed that Nash–Cournot equilibria of a network game converge to a Wardrop equilibrium as players split into replicas; Paccagnan et al. describe theirs as the only earlier bound on the distance between the equilibria.

The sharpest general statement we know is that of Paccagnan et al. (2019) for aggregative games with coupling constraints, in which a player's cost depends on its own strategy and the population average. Every Wardrop equilibrium is an ε-Nash equilibrium with $\varepsilon = O(1/M)$ (their Proposition 2). If the Nash or the Wardrop operator is strongly monotone in the individual strategies, the stacked strategy vectors satisfy $\|x^{\mathrm{N}} - x^{\mathrm{W}}\| \le L_2/(\alpha_M\sqrt{M})$ (Theorem 1(1)–(2)), which is of order $1/M$ for the population average when α_M stays bounded away from zero. These parts of their theorem therefore already give an $O(1/M)$ rate for games with strong monotonicity. If only the price map is strongly monotone, the averages satisfy $\|\sigma(x^{\mathrm{N}}) - \sigma(x^{\mathrm{W}})\| \le (2RL_2/(\alpha M))^{1/2}$ (Theorem 1(3)), of order $1/\sqrt{M}$. In our game profits are linear in the routing plan, the first condition fails and the second applies. Jacquot and Wan (2018) bound the squared distance between an atomic and the Wardrop equilibrium on parallel networks by terms linear in the mass of the largest class, again a square-root rate for the distance, and Cominetti et al. (2023) show that equilibrium flows of atomic congestion games with small weights approach Wardrop equilibria, with explicit rates.

Oligopoly theory asks the same question for Cournot markets: Novshek (1985) shows that equilibria of Cournot markets with shrinking firms converge to the competitive equilibrium of a market with infinitesimal firms. The potential structure we use is that of Slade (1994) and Monderer and Shapley (1996); in one zone with a slack fleet budget our game is a linear Cournot market and $Q^{\mathrm{N}} = \frac{N}{N+1}Q^{\mathrm{W}}$ is the textbook formula.

Relation to this paper. The rate itself is not new. Paccagnan et al. already give $O(1/M)$ under strong monotonicity in individual strategies and $O(1/\sqrt{M})$ for the aggregate otherwise, and our general bound (Theorem 4) uses their argument, the sum of two variational inequalities; the constant $\frac{1}{2}$ and the extension to pure equilibria with lumpy plans, through the Wardrop equilibrium of the mixed extension, are our additions. What we add is for games where strong monotonicity fails: the tightness family (Proposition 5), which, as far as we found, is the first lower bound showing that the square-root order is attained when profits are linear in routing plans that couple zones; the constant-one theorem for zone-separable carriers, sharp for identical carriers (Theorem 6); the bound for identical carriers under any coupling (Proposition 7); and the replica rate with an instance-dependent constant (Theorem 8), which rests on the upper-Lipschitz property of polyhedral multifunctions (Robinson, 1981).

2.2. *Mean-field games*

Mean-field games were introduced by Lasry and Lions (2007) and, with decentralised ε-Nash strategies, by Huang et al. (2006); a monograph is Carmona and Delarue (2018). The finite-state case is closest to ours. Gomes et al. (2010, 2013) study discrete- and continuous-time games on finite state spaces; to our reading of the abstracts, the convergence rate for N-player games that they state is for continuous time and short horizons. Doncel et al. (2019) prove convergence of N-player equilibria within a class of strategies and exhibit equilibria outside it that do not converge, and Cardaliaguet et al. (2019) prove convergence of the Nash system through the master equation. These results concern dynamics with Markov or smoothing structure and rates that depend on the regularity of the master-equation solution; they do not cover a static aggregative game with linear profits over polytopes, where the square-root rate appears.

Congestion enters through the cost. Guéant (2015) proves existence and a uniqueness criterion for congestion on graphs, and Al Saleh et al. (2024) relate first-order mean-field games

on networks to Wardrop equilibrium, so the link between mean-field games and Wardrop equilibrium is already established. Cabannes et al. (2021) solve N-player dynamic routing games through their limit and report accuracy beyond a few dozen vehicles, without a theorem on the gap. With a symmetric coupling the game is a potential mean-field game whose equilibria solve an optimisation problem, for which fictitious play (Cardaliaguet and Hadikhanloo, 2017) and the generalised conditional gradient (Lavigne and Pfeiffer, 2023) are solvers. Games with a major player and many minor ones (Huang, 2010) are the faithful model for a few large and many small carriers; this paper does not treat them.

Relation to this paper. Known: the potential structure, existence and uniqueness of the aggregate, solvers, and the Wardrop link on networks. We use occupation measures on a time-expanded graph, which make the carrier's problem a flow polytope (Calderone and Sastry, 2017), and the solver literature only as a model of decentralised adjustment (Section 8). New: a finite-N analysis for the Cournot-congestion coupling with lumpy multi-stop plans, which none of the convergence results above covers, and its transfer to the city's charge and welfare. We say nothing about rates for dynamic games with noise.

2.3. Road pricing, market power and Stackelberg games

Marginal-cost pricing goes back to Pigou (1920) and, on networks, to Beckmann et al. (1956); Yang and Huang (2005) give the theory for road networks. When the toll set is restricted the problem is second best (Verhoef, 2002), and when demand and cost functions are unknown, tolls can be learned from observed flows by trial-and-error procedures that converge to the marginal-cost toll (Yang et al., 2004). With market power among the users of a congested resource the optimal toll departs from the Pigou toll: Brueckner (2002) and Pels and Verhoef (2004) show this for airports served by Cournot carriers, and Buchanan (1969) is the classical reference for the dependence of corrective taxes on market structure. For routing by Markov decision processes, Calderone and Sastry (2017) show that nonatomic equilibria of MDP congestion games are Wardrop equilibria of a potential game. A leader who commits before a large population responds is studied in Stackelberg mean-field games: Guo et al. (2022) reformulate a leader who maximises against the worst ε-Nash response of the followers as a minimax problem, and Sanjari et al. (2022) show that with finitely many followers symmetric incentive strategies exist that are approximately optimal for the leader, whereas with an infinite population a smooth finite-energy incentive strategy can fail to exist.

Relation to this paper. Known: marginal-cost tolls implement the optimum in the limit; trial-and-error tolls converge without demand information; with Cournot players the optimal toll is the Pigou toll corrected by a market-power term. Proposition 20 is a finite-N analogue of these second-best and market-structure results (Buchanan, 1969; Brueckner, 2002; Pels and Verhoef, 2004); its only novelty is the closed form for identical carriers with multi-stop plans under an arbitrary coupling of zone-hours, and its economic content (a subsidy when $N < m_k/\kappa_k$) is that of the airport literature. Proposition 11(a) restates the contraction theorem of the companion paper (Dang, 2026), a trial-and-error rule of the kind above. New: the exact identity for the finite-market fixed point of the observation-only rule, κ times the Nash aggregate of an auxiliary game (Theorem 12), and the uniform $O(\bar{s})$ welfare bound for that rule (Theorem 13(b)). The city here posts prices of zone-hour aggregates from observed totals; we do not address the existence problem of smooth incentives raised by Sanjari et al. (2022).

2.4. Urban logistics, delivery fleets and ride-hailing

Cattaruzza et al. (2017) review the routing problems of city logistics, and evidence on charging comes from London (Leape, 2006). Such studies evaluate a given policy on given traffic; they do not model carriers who choose multi-stop tours in response to zone-time charges. Gig

drivers are documented by Hall and Krueger (2018), and Arslan et al. (2019) model crowd-sourced delivery. Equilibrium models of ride-hailing include the mean-field model of Li et al. (2025) with repositioning across regions; to our reading of the abstracts, such models have no congestion network with zone-time charges and no multi-stop routing. The companion paper (Dang, 2026) studies a city that charges a handful of large carriers who solve routing problems with probability-box travel times.

Relation to this paper. New relative to these studies: a strategic model in which many small carriers with multi-stop tours and fleet budgets respond to zone-hour charges on a time-expanded graph, with the convergence of the finite market. We reuse the model, calibration and nine cities of the companion paper and replace its atomic setting by the many-small-carrier limit.

2.5. Robust games, entry and external costs

Gilboa and Schmeidler (1989) axiomatise maxmin expected utility over a set of priors and Iyengar (2005) solves Markov decision processes with rectangular uncertainty sets; for finite-state mean-field games with ambiguity in transition probabilities, Langner et al. (2024) prove existence of an equilibrium. We use only the Γ-maximin surcharge of the companion paper, which in the limit acts like a per-delivery tax (Proposition 15), and we do not contribute to the theory of robust mean-field games.

Free entry in Cournot markets can be socially excessive when entrants steal business from incumbents (Mankiw and Whinston, 1986), and optimal taxes under free entry depart from the Pigou level by a term whose sign depends on the curvature of demand and cost (Konishi et al., 1990; Katsoulacos and Xepapadeas, 1995; Hamada et al., 2022). In transport, Brueckner (2002) and Pels and Verhoef (2004) hold the number of carriers fixed. External costs of transport are surveyed for the European Union in the van Essen et al. (2020), and distributional effects of road pricing are analysed by Eliasson and Mattsson (2006). Proposition 22(c) is the business-stealing result of Mankiw and Whinston; what is new is limited to the identity at zero profit (part (a)), the one-zone linear second-best result (part (d)) and the numerical findings of Section 12.

2.6. What is new

To our reading, the following are new; the rest is known and used as such (potentials and uniqueness, marginal-cost tolls, MDP congestion games, the contraction of the observation-only rule, robust-MFG existence, free-entry accounting).

1. For games where strong monotonicity in individual strategies fails, a lower bound for the Nash–Wardrop distance: a two-zone family whose aggregate gap attains the general bound up to the factor $1/\sqrt{2}$ (Proposition 5). The square-root order is thus the exact worst-case order once plans couple zones, whereas Paccagnan et al. (2019) already give $O(1/M)$ under strong monotonicity.
2. A constant-one theorem for zone-separable carriers, convexified or lumpy, with equality for identical carriers (Theorem 6); the bound $\|A^{\mathrm{N}}\|_m/N$ for identical carriers under any coupling (Proposition 7); and $O(1/n)$ for replicas with an instance-dependent constant (Theorem 8, Proposition 9).
3. An exact identity for the finite-market fixed point of the observation-only charge (Theorem 12) and a welfare loss of at most $\frac{1}{2}\bar{s}\langle m, Q^{\mathrm{opt}}\rangle$ for it, for every instance (Theorem 13(b)).
4. The closed form of the optimal charge for N identical carriers under arbitrary plan coupling, Pigou minus $\frac{1}{N} m \odot Q^{\mathrm{opt}}$ (Proposition 20), a finite-N analogue of known second-best results; and the fee, free-entry and tour-externality propositions (Propositions 21–23), short results in known frameworks that organise the computations of Sections 11 and 12.
5. Findings on nine cities with a stylised hourly profile: the gap is about one carrier's delivery throughout; the observation-only charge lowers welfare with few carriers; the market-power subsidy attracts entrants; through-traffic delay dominates the external cost of tours.

Items 1–3 use the argument of Paccagnan et al. (2019), and the replica rate rests on Robinson (1981); item 4 and the contraction of the observation-only rule are close to Brueckner (2002), Pels and Verhoef (2004), Yang et al. (2004) and the companion paper.

3. Model

3.1. The time-expanded zone graph

The city is divided into zones $z \in [Z]$ and the delivery day into periods $t \in [H]$ (in the experiments $H = 10$ hourly periods from 08:00 to 18:00). The *time-expanded zone graph* G_H has a node (z, t) for every zone and period, a travel arc $(z, t) \to (z', t')$ whenever a vehicle that leaves z in period t reaches z' in period $t' \geq t$ (travel times depend on the departure period), and a waiting arc $(z, t) \to (z, t + 1)$. Because every arc moves forward in time, G_H is acyclic.

A *tour* of a carrier is a path in G_H that starts and ends at the carrier's depot and makes deliveries at some of the nodes it visits. Delivering in zone z during period t adds to the deliveries q_{zt} of the carrier at that node. The coordinates of the interaction are the nodes $k = (z, t)$ touched by some tour; we write $\mathcal{K}$ for this set and $d = |\mathcal{K}| \leq ZH$. Vehicle-hours spent in zone z during period t are $r\, q_{zt}$, where r is the in-zone time per delivery (assumption C3 of Dang, 2026): congestion is caused by the time spent serving a zone, and inter-zone driving is priced through the tour's own cost only. This is the delivery-unit reduction of the companion paper.

Demand and congestion. In coordinate k, the inverse demand for deliveries is $\alpha_k - b_k Q_k$ with $b_k > 0$, where Q_k is the total number of deliveries made at k by all carriers. A vehicle-hour at k costs $\gamma(\xi_k + c_k X_k)$ where $X_k = rQ_k$ is the congestion variable. Per delivery, a carrier therefore pays $\gamma r \xi_k$ (part of its private cost) plus $\kappa_k Q_k$ with $\kappa_k = \gamma c_k r^2 > 0$. Write

$$m_k = b_k + \kappa_k, \qquad M = \operatorname{diag}(m), \qquad \varrho = \max_k \kappa_k / m_k < 1.$$

For $v \in \mathbb{R}^d$ and a positive weight vector w we write $\|v\|_w^2 = \sum_k w_k v_k^2$; thus $\|v\|_m$ is the M-norm and $\|v\|_{m^{-1}}$ its dual norm.

3.2. Carriers, plans and their convexification

There are N carriers. Carrier i has a finite set $\mathcal{P}_i$ of *plans*: a plan is a multiset of tours that the carrier's fleet can operate in one day. Plan p has deliveries $q_i(p) \in \mathbb{R}^d_{\geq 0}$ and private profit $\pi_i(p) = \sum_k \alpha_k q_{ik}(p) - C_i(p)$, where $C_i(p)$ is its routing cost (driver time, fuel, fixed costs and the free-flow congestion term $\gamma r \langle \xi, q_i(p) \rangle$). A *small multi-stop carrier* is a carrier whose fleet is small compared with the market; the mean-field limit lets the share of every carrier go to zero while plans keep their routing structure.

For most results the plan set is convexified. A carrier with a fleet budget B_i (vehicle-hours) and a tour set $\mathcal{T}_i$ chooses fleet weights $y_i \in Y_i = \{y \in \mathbb{R}^{\mathcal{T}_i}_{\geq 0} : \langle w_i, y \rangle \leq B_i\}$, where w_{ij} is the duration of tour j; its deliveries are $a_i = L_i y_i$ with $L_i \geq 0$ entrywise (column j is the delivery vector of tour j per vehicle-hour) and its profit is $\pi_i(y_i) = \langle p_i, y_i \rangle$. The abstract assumptions under which the theory holds are:

(A1) Y_i is a nonempty compact convex subset of a Euclidean space, $\pi_i : Y_i \to \mathbb{R}$ is concave and upper semicontinuous, and $a_i = L_i y_i$ with L_i linear and $L_i y_i \in \mathbb{R}^d_{\geq 0}$ for $y_i \in Y_i$;

(A2) $m_k > 0$ for all k.

The fleet polytope with linear profits satisfies (A1). We write $A = \sum_i a_i$ (in delivery units $A = Q$) and

$$\bar{s} = \max_i \max_{y \in Y_i} \max_k (L_i y)_k$$

for the largest number of deliveries that one carrier can make at one coordinate; $\bar{s}$ is the *share* parameter of the limit.

Payoffs and equilibria. The city posts a charge $u \in \mathbb{R}^d$ per delivery. Carrier i's payoff is

$$J_i(y_i; y_{-i}) = \pi_i(y_i) - \langle a_i, M(A_{-i} + a_i)\rangle - \langle u, a_i\rangle, \qquad A_{-i} = \sum_{j \neq i} a_j. \tag{1}$$

A *Nash equilibrium* y^{N} is a profile in which each y_i maximises $J_i(\cdot\,; y_{-i})$: a carrier takes its own effect on the price and the congestion of the coordinates it serves into account (Cournot behaviour). A *Wardrop (mean-field) equilibrium* y^{W} is a profile in which each y_i maximises $\pi_i(y_i) - \langle a_i, MA + u\rangle$ with the aggregate $A = A(y^{\mathrm{W}})$ held fixed: carriers are price takers. The mean-field reading is the usual one: if carrier i is replaced by a continuum of identical infinitesimal agents whose plans sum to a point of Y_i, each agent's own effect vanishes and the equilibrium conditions of the continuum are those of y^{W}. On G_H this is the Wardrop equilibrium of a Markov-decision-process congestion game in which vehicles choose paths (Calderone and Sastry, 2017), with an elastic (Cournot) price added; it is also a discrete-time, finite-state potential mean-field game (Gomes et al., 2010; Guéant, 2015).

The welfare of a profile, with charges treated as transfers, is

$$\mathrm{SW}(y) = \sum_i \pi_i(y_i) + \sum_k \Big[-\tfrac{b_k}{2} Q_k^2 - \kappa_k Q_k^2 \Big] = \sum_i \pi_i(y_i) - \tfrac{1}{2}\|Q\|^2_{m+\kappa}, \tag{2}$$

the gross consumer benefit $\alpha_k Q_k - \frac{b_k}{2} Q_k^2$ (its linear part is inside π_i) minus routing and congestion costs. A *social optimum* maximises SW.

3.3. Potentials and a linearisation lemma

The first lemma is elementary convex analysis; it is used in almost every proof below.

Lemma 1 (Linearisation)**.** *Let C be convex, $h : C \to \mathbb{R}$ concave and g convex and differentiable on a neighbourhood of C. If x^* maximises $h - g$ over C, then x^* maximises $h - \langle \nabla g(x^*), \cdot\rangle$ over C.*

Proof. Let $x \in C$ and $x_t = x^* + t(x - x^*)$ for $t \in (0, 1]$. Optimality and concavity of h give $0 \geq h(x_t) - g(x_t) - h(x^*) + g(x^*) \geq t\,(h(x) - h(x^*)) - (g(x_t) - g(x^*))$. Divide by t and let $t \downarrow 0$: $0 \geq h(x) - h(x^*) - \langle \nabla g(x^*), x - x^*\rangle$. □

Lemma 2 (Potentials, uniqueness and the own surcharge)**.** *Assume (A1)–(A2) and fix u. Let*

$$\Phi_{\mathrm{N}}(y) = \sum_i \pi_i(y_i) - \langle u, A\rangle - \tfrac{1}{2}\|A\|^2_m - \tfrac{1}{2}\sum_i \|a_i\|^2_m, \qquad \Phi_{\mathrm{W}}(y) = \sum_i \pi_i(y_i) - \langle u, A\rangle - \tfrac{1}{2}\|A\|^2_m.$$

(a) y is a Nash equilibrium iff it maximises Φ_{N} over $\prod_i Y_i$, and a Wardrop equilibrium iff it maximises Φ_{W}; a social optimum maximises Φ_{W} with $u = 0$ and m replaced by $m + \kappa$. All three sets are nonempty.

(b) The Nash contributions a_i^{N} are unique, and so is the Wardrop aggregate A^{W}; the Wardrop contributions need not be.

(c) (Own surcharge.) Every Nash equilibrium is a Wardrop equilibrium of the game in which carrier i pays the additional per-delivery charge $\sigma_i = Ma_i^{\mathrm{N}}$ (with a_i^{N} held fixed): each y_i^{N} maximises $\pi_i(y_i) - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle$.

Proof. (a) $\Phi_{\mathrm{N}} - J_i$ does not depend on y_i: the terms of Φ_{N} containing y_i are $\pi_i - \langle u, a_i\rangle - \langle A_{-i}, Ma_i\rangle - \|a_i\|^2_m$, which are those of J_i. Hence a maximiser of Φ_{N} is a Nash equilibrium. Conversely let y be a Nash equilibrium and write $\Phi_{\mathrm{N}} = \sum_i \pi_i - f$ with f convex and differentiable. By Lemma 1 applied to carrier i (with $h = \pi_i$, $g = f(\cdot, y_{-i})$), $\pi_i(y_i') - \langle \nabla_i f(y), y_i' - y_i\rangle \leq \pi_i(y_i)$ for all $y_i' \in Y_i$. Summing over i and using $f(y') \geq f(y) + \langle \nabla f(y), y' - y\rangle$ gives $\Phi_{\mathrm{N}}(y') \leq \Phi_{\mathrm{N}}(y)$. The Wardrop case is the same with the linear functional $y_i \mapsto \langle a_i, MA + u\rangle$, whose y_i-gradient is $\nabla_i(\frac{1}{2}\|A\|^2_m + \langle u, A\rangle)$. Existence: the potentials are upper semicontinuous on a compact set.

(b) Φ_{N} is the sum of a concave function and $-\frac{1}{2}\|A\|_m^2 - \frac{1}{2}\sum_i \|a_i\|_m^2$, which is strictly concave in $(a_1, \ldots, a_N)$; two maximisers with different contributions would have a midpoint with a strictly larger value. The same argument with $-\frac{1}{2}\|A\|_m^2$ gives uniqueness of A^{W}. Non-uniqueness of Wardrop contributions: two identical carriers with linear profits can exchange deliveries. (c) Lemma 1 applied to J_i with $g(y_i) = \|L_i y_i\|_m^2$, whose gradient at y_i^{N} is $2L_i^\top M a_i^{\mathrm{N}}$, and $A_{-i} + a_i^{\mathrm{N}} = A^{\mathrm{N}}$. □

Parts (a) and (b) are classical: the Nash potential of a Cournot game with linear demand is due to Slade (1994) and the Wardrop potential to Beckmann et al. (1956); for MDP congestion games see Calderone and Sastry (2017). Part (c) is the device behind all the bounds below: the finite game is the mean-field game in which each carrier pays a private surcharge equal to its own marginal effect Ma_i^{N}, a vector of size at most $\max_k m_k \cdot \bar{s}$.

Replicas. A *base type* θ is a triple $(\mathcal{Y}_\theta, \pi_\theta, L_\theta)$ satisfying (A1) with π_θ positively homogeneous (linear profits are). Its *n-split* consists of n carriers with $Y = \mathcal{Y}_\theta/n$, profit π_θ and the same L_θ: the fleet of the type is divided equally. A game whose carriers are the n_θ-splits of base types $\theta \in \Theta$ is a *replica game*.

Lemma 3 (Symmetry of replicas)**.** *In a replica game, carriers of the same type have the same Nash contributions, and the type totals $Y_\theta = \sum_{i\in\theta} y_i$ of a Nash equilibrium maximise*

$$\Phi^{(n)}(Y) = \sum_\theta \pi_\theta(Y_\theta) - \langle u, A\rangle - \tfrac{1}{2}\|A\|_m^2 - \tfrac{1}{2}\sum_\theta \frac{1}{n_\theta}\|L_\theta Y_\theta\|_m^2, \qquad A = \sum_\theta L_\theta Y_\theta,$$

over $\prod_\theta \mathcal{Y}_\theta$. The Wardrop aggregate does not depend on (n_θ).

Proof. Φ_{N} is invariant under permutations of identical carriers, so by uniqueness of the Nash contributions (Lemma 2(b)) identical carriers have equal contributions. Averaging a maximiser over the permutations within each type gives a symmetric maximiser (concavity), whose value is $\Phi^{(n)}$ of its type totals; conversely every Y yields the symmetric profile $y_i = Y_\theta/n_\theta$. For the Wardrop part, Φ_{W} depends on y only through the type totals, and $\sum_{i\in\theta} \mathcal{Y}_\theta/n_\theta = \mathcal{Y}_\theta$ by convexity. □

Lemma 3 is also the computational reduction used in Section 8: the Nash equilibrium of n replicas per type costs one convex quadratic program in the type totals, whatever n is.

3.4. Imprecise beliefs and vehicle counts

Two extensions of the companion model are kept. Under Γ-maximin beliefs of width δ about the rival aggregate (companion paper, Lemma "Γ-maximin surcharge") each carrier evaluates its plans at the upper corner $A_{-i} + \delta w$ of a box of weights $w \in \mathbb{R}^d_{>0}$; Section 6 studies this layer in the limit. Under vehicle-count congestion (D2) the congestion variable counts vehicles instead of vehicle-hours; Section 7 treats it.

4. How far is the finite market from its mean-field limit?

Throughout this section the charge u is fixed (all bounds hold uniformly in u), y^{N} is a Nash equilibrium and y^{W} a Wardrop equilibrium of the same carriers, and $\Delta A = A^{\mathrm{N}} - A^{\mathrm{W}}$.

4.1. A general bound and its tightness

Theorem 4 (General bound)**.** *Assume (A1)–(A2).*

(a) For every Wardrop equilibrium y^{W},

$$\|\Delta A\|_m^2 \le \sum_i \langle a_i^{\mathrm{N}}, M(a_i^{\mathrm{W}} - a_i^{\mathrm{N}})\rangle \le \frac{1}{4}\sum_i \|a_i^{\mathrm{W}}\|_m^2,$$

hence

$$\|\Delta A\|_m \le \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|_m^2\Big)^{1/2} \le \frac{1}{2}\big(\bar{s}\,\langle m, A^{\mathrm{W}}\rangle\big)^{1/2}.$$

(b) (Lumpy plans.) Let p be a pure Nash equilibrium of the game in which carrier i chooses a plan in the finite set $\mathcal{P}_i$ (payoff (1) with $a_i = q_i(p_i)$), and let x^{W} be a Wardrop equilibrium of its mixed extension (carrier i mixes over $\mathcal{P}_i$, contributes the mean $\bar{a}_i$). Then

$$\|A(p) - A^{\mathrm{W}}\|_m^2 \le \sum_i \Big(\mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2 - \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle\Big) \le \bar{s}\,\langle m, A^{\mathrm{W}}\rangle,$$

with $\bar{s}$ the largest delivery of a plan at a coordinate. A pure Nash equilibrium exists.

Proof. (a) By Lemma 2(c), y_i^{N} maximises $\pi_i - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle$, so it is at least as good as y_i^{W}; and y_i^{W} maximises $\pi_i - \langle a_i, MA^{\mathrm{W}} + u\rangle$, so it is at least as good as y_i^{N}:

$$\pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle \le \pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle,$$

$$\pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{W}} + u\rangle \le \pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{W}} + u\rangle.$$

Adding, the profits and charges cancel and $\langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \le \langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, Ma_i^{\mathrm{N}}\rangle$. Summing over i gives the first inequality. The second is $\langle x, M(y-x)\rangle \le \frac{1}{4}\|y\|_m^2$, i.e. $\|x - y/2\|_m^2 \ge 0$, with $x = a_i^{\mathrm{N}}$, $y = a_i^{\mathrm{W}}$. Finally $\sum_i \|a_i^{\mathrm{W}}\|_m^2 = \sum_k m_k \sum_i (a_{ik}^{\mathrm{W}})^2 \le \bar{s}\sum_k m_k A_k^{\mathrm{W}}$ because $0 \le a_{ik}^{\mathrm{W}} \le \bar{s}$.

(b) A pure equilibrium exists because Φ_{N} restricted to pure profiles is an exact potential of a finite game (Monderer and Shapley, 1996). At p, carrier i does not gain by deviating to any plan p'; averaging over $p' \sim x_i^{\mathrm{W}}$ (payoffs are evaluated against the realised rival aggregate $A_{-i} = A(p) - q_i(p_i)$),

$$\pi_i(p_i) - \langle q_i(p_i), M(A_{-i} + q_i(p_i))\rangle \ge \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle \bar{a}_i^{\mathrm{W}}, MA_{-i}\rangle - \mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2.$$

The Wardrop condition, tested against the pure strategies p, gives $\sum_i \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle A^{\mathrm{W}}, MA^{\mathrm{W}}\rangle \ge \sum_i \pi_i(p_i) - \langle A(p), MA^{\mathrm{W}}\rangle$ (the charge terms are omitted; they cancel as in (a)). Adding the sum over i of the first inequality to the second and expanding A_{-i} gives $0 \ge \|A(p) - A^{\mathrm{W}}\|_m^2 + \sum_i \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle - \sum_i \mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2$, which is the claim; the last bound uses $\langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle \ge 0$ and $\mathbb{E}\|q_i\|_m^2 \le \bar{s}\,\langle m, \bar{a}_i^{\mathrm{W}}\rangle$. □

The first inequality of (a) is the summed variational-inequality argument of Paccagnan et al. (2019, Thm. 1(3)), written for this game; with $\bar{s} \sim 1/N$ it gives the rate $O(N^{-1/2})$ of that theorem. Under strong monotonicity in the individual strategies, their Theorem 1(1)–(2) already give $O(1/N)$; that condition fails here because profits are linear in the plans. The constant $\frac{1}{2}$ and part (b) are additions. The next result shows that, without further structure, nothing better than $O(\bar{s}^{1/2})$ holds: the constant $\frac{1}{2}$ is attained up to the factor $1/\sqrt{2}$.

Proposition 5 (Tightness)**.** *Let $n \ge 2$ and $\varphi_n \in (0, \pi/2)$ with $1 - \cos\varphi_n = 1/n$; put $c = \cos\varphi_n$, $v = (c, \sin\varphi_n)$. Consider two coordinates with $M = I$ and $2n$ carriers: n carriers of type A choose $t \in [0, 1/n]$, deliver te_1 and earn $\alpha_A t$; n carriers of type B choose $t \in [0, 1/n]$, deliver tv and earn $\alpha_B t$; $(\alpha_A, \alpha_B) = (1, c)$. Then the Wardrop aggregate is $A^{\mathrm{W}} = e_1$ (type totals $T^{\mathrm{W}} = (1, 0)$), the Nash type totals are $T^{\mathrm{N}} = (\frac{3}{4} - \frac{1}{4n}, \frac{1}{4} - \frac{1}{4n})$, and*

$$\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_2^2 = \frac{1}{8n} + \frac{2n-1}{8n^3}, \qquad \text{so} \qquad \|\Delta A\|_2 \ge \frac{1}{2\sqrt{2n}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|^2\Big)^{1/2}.$$

Every carrier can deliver at most $\bar{s} = 1/n$ at any coordinate, while $\|\Delta A\|_\infty \geq \|\Delta A\|_2/\sqrt{2} \geq \frac{1}{4}\sqrt{n}\,\bar{s}$. Hence there is no constant C, independent of the instance, with $\|A^{\mathrm{N}} - A^{\mathrm{W}}\| \leq C\bar{s}$, even with two coordinates, linear profits and bounded data; over this class the worst-case gap is of exact order $\bar{s}^{1/2}$.

Proof. Write $J = [e_1, v]$ and $G = J^\top J = \left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right) \succ 0$. By Lemma 3 it suffices to work with the type totals $T \in [0,1]^2$, $A = JT$. The Wardrop potential is $\langle \alpha, T\rangle - \frac{1}{2}T^\top GT$; its unconstrained maximiser $G^{-1}\alpha = (1,0)$ lies in the box (on its boundary, where both partial derivatives $\alpha - GT$ vanish), so $T^{\mathrm{W}} = (1,0)$ is the maximiser; it is unique because $G \succ 0$. The Nash potential of Lemma 3 is $\langle \alpha, T\rangle - \frac{1}{2}T^\top(G + \frac{1}{n}I)T$ (each type moves along a unit vector, so $\|L_\theta T_\theta\|^2 = T_\theta^2$); its unconstrained maximiser solves $(G + \frac{1}{n}I)T = \alpha$, and direct substitution (with $c = 1 - 1/n$) shows that T^{N} above solves it; it lies in the interior of the box, so it is the Nash equilibrium. Then $\Delta A = J(T^{\mathrm{N}} - T^{\mathrm{W}})$ with $T^{\mathrm{N}} - T^{\mathrm{W}} = -(\frac{1}{4} + \frac{1}{4n}, -\frac{1}{4} + \frac{1}{4n})$, and $\|\Delta A\|^2 = \Delta T^\top G \Delta T$ evaluates (using $\sin^2\varphi_n = 1 - c^2$) to the stated expression, which is at least $1/(8n)$. With the Wardrop selection in which every A carrier delivers $1/n$ and every B carrier nothing, $\sum_i \|a_i^{\mathrm{W}}\|^2 = 1/n$. Per-carrier deliveries are at most $1/n$ by construction. □

The mechanism is visible in the eigenvectors of G: when two types move along nearly parallel directions (φ small), their *difference* direction $e_1 - v$ is almost free for the mean-field market, so a small difference between the types' own surcharges (Ma_i^{N}, of size $1/n$) moves the type totals by $O(1/(n\lambda_{\min}(G)))$ in opposite directions, and the aggregate by $O(1/(n\lambda_{\min}(G)^{1/2}))$. For a *fixed* angle φ the same computation gives $n\|\Delta A\|_2 \to 1/\sin\varphi$ as $n \to \infty$: the rate is $O(1/n)$, but with a constant that grows without bound as the types become collinear (Proposition 9 below states this in general). Proposition 5 shows that an instance-free constant for the cross-zone case does not exist.

4.2. Exact $O(\bar{s})$ bounds under structure

Theorem 6 (Zone-separable carriers: constant one)**.** *Suppose every carrier's problem separates by coordinate: $Y_i = \prod_k Y_{ik}$, $\pi_i = \sum_k \pi_{ik}$, and a_{ik} depends on y_{ik} only, so that carrier i chooses a number of deliveries q_{ik} in a compact set $S_{ik} \subset \mathbb{R}_{\geq 0}$ (an interval for the convexified game, a finite set for lumpy plans) with concave (respectively arbitrary) profit π_{ik}. Then for every coordinate k and every charge u:*

(a) (convexified game) $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i q_{ik}^{\mathrm{N}}$;

(b) (lumpy plans) every pure Nash equilibrium satisfies $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i \max S_{ik}$, where Q^{W} is the Wardrop aggregate of the mixed extension.

In (a) equality holds when the N carriers are identical with linear profit and an interior equilibrium: then $Q_k^{\mathrm{N}} = \frac{N}{N+1}Q_k^{\mathrm{W}}$ and $Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} = Q_k^{\mathrm{W}}/(N+1) = q_{ik}^{\mathrm{N}}$.

Proof. Separability decouples the coordinates, so fix k, drop it from the notation and write $m = m_k$, $\pi_i = \pi_{ik}$, $Q = Q^{\mathrm{N}}$ and $q_i = q_i^{\mathrm{N}}$; the charge u_k is absorbed into π_i. Let $\hat{s}_i = q_i$ in case (a) and $\hat{s}_i = \max S_i$ in case (b). We first show two revealed-preference facts about the Nash choice q_i:

(i) for all $q' \in S_i$ with $q' < q_i$: $\pi_i(q_i) - mQq_i \geq \pi_i(q') - mQq'$;

(ii) for all $q' \in S_i$ with $q' > q_i$: $\pi_i(q_i) - m(Q + \hat{s}_i)q_i \geq \pi_i(q') - m(Q + \hat{s}_i)q'$.

In case (a), Lemma 2(c) says that q_i maximises $\pi_i(q) - m(Q + q_i)q$ over S_i, which implies (ii) directly and (i) because $m(Q + q_i)(q_i - q') \geq mQ(q_i - q')$ for $q' < q_i$. In case (b), the Nash condition against q' reads $\pi_i(q_i) - \pi_i(q') \geq m[(q_i - q')Q_{-i} + q_i^2 - q'^2] = m(q_i - q')(Q_{-i} + q_i + q')$ with $Q_{-i} = Q - q_i$; for $q' < q_i$ the right side is at least $m(q_i - q')Q$, and for $q' > q_i$ it is at least $-m(q' - q_i)(Q + \hat{s}_i)$ because $q' \leq \hat{s}_i$.

Now let $p \geq 0$ and let $\mathrm{BR}_i(p) = \arg\max_{q \in S_i} \pi_i(q) - pq$; in case (b) a Wardrop carrier mixes over $\mathrm{BR}_i(p)$, in case (a) it chooses a point of the interval $\mathrm{BR}_i(p)$; either way its delivery lies in $\operatorname{conv} \mathrm{BR}_i(p)$ and $Q^{\mathrm{W}} = \sum_i \bar{q}_i^{\mathrm{W}}$ with $\bar{q}_i^{\mathrm{W}} \in \operatorname{conv} \mathrm{BR}_i(mQ^{\mathrm{W}})$.

Upper bound $Q \leq Q^{\mathrm{W}}$. Suppose $Q > Q^{\mathrm{W}}$, so $p := mQ^{\mathrm{W}} < mQ$. For $q' < q_i$, (i) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (mQ - p)(q_i - q') > 0$, so no $q' < q_i$ is a best response at p, and $\bar{q}_i^{\mathrm{W}} \geq q_i$ for every i. Summing, $Q^{\mathrm{W}} \geq Q$, a contradiction.

Lower bound $Q \geq Q^{\mathrm{W}} - \hat{s}$, $\hat{s} = \max_i \hat{s}_i$. Suppose $Q < Q^{\mathrm{W}} - \hat{s}$, so $p := mQ^{\mathrm{W}} > m(Q + \hat{s}) \geq m(Q + \hat{s}_i)$. For $q' > q_i$, (ii) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (p - m(Q + \hat{s}_i))(q' - q_i) > 0$, so $\bar{q}_i^{\mathrm{W}} \leq q_i$ for all i and $Q^{\mathrm{W}} \leq Q$, a contradiction.

Equality case: with profit αq and an interior solution, the first-order conditions are $\alpha = mQ^{\mathrm{W}}$ and $\alpha = mQ^{\mathrm{N}} + mq$ with $Q^{\mathrm{N}} = Nq$, so $q = \alpha/(m(N+1))$ and $Q^{\mathrm{W}} - Q^{\mathrm{N}} = \alpha/m - N\alpha/(m(N+1)) = q$. □

The theorem needs no symmetry, no smoothness and no bound on the number of carriers, and the constant is attained. Its scope is a carrier whose plans in different zone-hours do not compete for the same fleet. A carrier whose fleet budget is shared across zones, or whose tours visit several zones, is not separable; Proposition 5 shows that for such carriers no analogue with an instance-free constant holds.

Proposition 7 (Equal own surcharges)**.** *If all carriers have the same Nash contribution $\bar{a}$ (in particular, if they are identical: Lemma 3), then $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \leq \|\bar{a}\|_m = \|A^{\mathrm{N}}\|_m/N$, for any coupling across coordinates.*

Proof. In the first inequality of Theorem 4(a), $\sum_i \langle \bar{a}, M(a_i^{\mathrm{W}} - \bar{a})\rangle = \langle \bar{a}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \leq \|\bar{a}\|_m \|\Delta A\|_m$. □

Theorem 8 (Replicas: $O(1/n)$ with an instance-dependent constant)**.** *Consider a replica game with base types $\theta \in \Theta$, polyhedral $\mathcal{Y}_\theta$ and linear profits, and let $n = \min_\theta n_\theta$. There is a constant C, depending on the base instance (and on u) but not on (n_θ), such that $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \leq C/n$ for all $n \geq 1$.*

Proof. For $\sigma = (\sigma_\theta)_\theta \in \mathbb{R}^{d \times \Theta}$ let $S(\sigma)$ be the Wardrop aggregate of the base types (one carrier per type) when type θ pays the extra per-delivery charge σ_θ; it is unique by Lemma 2(b), and $S(0) = A^{\mathrm{W}}$ for every replica game (Lemma 3). By Lemmas 3 and 2(c), the Nash aggregate of the replica game is $S(\sigma^n)$ with $\sigma_\theta^n = ML_\theta Y_\theta^{\mathrm{N}}/n_\theta$, so $\|\sigma^n\| \leq c_1/n$ with $c_1 = \max_\theta \max_{Y \in \mathcal{Y}_\theta} \|ML_\theta Y\|$. The Wardrop equilibria under σ are the solutions of the Karush–Kuhn–Tucker system of a convex quadratic program with polyhedral constraints in which σ enters linearly; its solution set, as a multifunction of σ, has a graph that is a finite union of polyhedra. By Robinson (1981) such a multifunction is locally upper Lipschitz: there are $\eta, C_0 > 0$ with $S(\sigma) \in S(0) + C_0\|\sigma\|\mathbb{B}$ whenever $\|\sigma\| \leq \eta$. Hence $\|\Delta A\| \leq C_0 c_1/n$ for $n \geq c_1/\eta$, and $\|\Delta A\| \leq D \leq D(c_1/\eta)/n$ otherwise, where D is the diameter of the set of feasible aggregates. □

The constant of Theorem 8 is not explicit. In the smooth interior case it is:

Proposition 9 (Affine interior replicas: explicit constant)**.** *Let base type θ choose t_θ in an open set, deliver $L_\theta t_\theta$ and earn $\langle c_\theta, t_\theta\rangle - \frac{1}{2} t_\theta^\top K_\theta t_\theta$ with $K_\theta \succeq 0$ (its n_θ-split carriers earn $\langle c_\theta, t\rangle - \frac{n_\theta}{2} t^\top K_\theta t$). Let $L = [L_\theta]_\theta$, $G = L^\top M L$, $K = \operatorname{diag}(K_\theta)$, $D_n = \operatorname{diag}(L_\theta^\top M L_\theta/n_\theta)$, and suppose $G + K \succ 0$ and that both equilibria are interior. Then*

$$\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m = \|G^{1/2}(G + K + D_n)^{-1} D_n T^{\mathrm{W}}\|_2 \leq \|D_n T^{\mathrm{W}}\|_{(G+K)^{-1}}$$
$$\leq \lambda_{\min}(G + K)^{-1/2} \Big(\sum_\theta \|L_\theta^\top M a_\theta^{\mathrm{W}}\|^2 \Big)^{1/2},$$

with $a_\theta^{\mathrm{W}} = L_\theta T_\theta^{\mathrm{W}}/n_\theta$ the per-carrier Wardrop delivery of type θ.

Proof. The first-order conditions are $(G+K)T^{\mathrm{W}} = c - L^{\top}u$ and $(G+K+D_n)T^{\mathrm{N}} = c - L^{\top}u$, so $\Delta T = -(G+K+D_n)^{-1}D_nT^{\mathrm{W}}$ and $\|\Delta A\|_m^2 = \Delta T^{\top}G\Delta T$. With $S = G+K+D_n \succeq G+K \succ 0$, $S^{-1}GS^{-1} \preceq S^{-1} \preceq (G+K)^{-1}$, which gives the first inequality; the second is $\|x\|_{(G+K)^{-1}} \le \lambda_{\min}(G+K)^{-1/2}\|x\|_2$ and $(D_nT^{\mathrm{W}})_\theta = L_\theta^{\top}Ma_\theta^{\mathrm{W}}$. □

When the profit of each type is linear ($K = 0$), the constant is $\lambda_{\min}(G)^{-1/2}$: the gap is $O(1/n)$ exactly when the types' delivery directions are linearly independent in the M-geometry, and the constant degrades as they approach collinearity, as in Proposition 5 where $\lambda_{\min}(G) = 1 - \cos\varphi$. With strictly concave individual profits ($K \succ 0$) the constant is at most $\lambda_{\min}(K)^{-1/2}$; this is the regime of Paccagnan et al. (2019, Thm. 1(1)–(2)), whose $O(1/N)$ bounds use strong monotonicity of the Nash or Wardrop operator in the individual strategies. Routing plans with linear costs and fleet budgets do not provide such monotonicity, which is why Theorem 8 has to rely on polyhedrality.

Summary of the rate question. With $\bar{s}$ the largest share of one carrier at one coordinate: the gap is at most $\frac{1}{2}(\bar{s}\langle m, A^{\mathrm{W}}\rangle)^{1/2}$ always (Theorem 4), and this order is attained (Proposition 5); it is at most $\bar{s}$ per coordinate with constant one for zone-separable carriers (Theorem 6) and at most $\|A^{\mathrm{N}}\|_m/N$ for identical carriers (Proposition 7); for replicas of a fixed instance it is $O(1/n)$ with a constant that depends on the conditioning of the instance (Theorem 8, Proposition 9).

5. The city's charge in the limit

The city posts a per-delivery charge $u \in \mathbb{R}^d$ (by the proposition "one instrument per zone" of the companion paper it can be realised as a uniform per-delivery charge plus non-negative congestion charges per vehicle-hour). Let $Q^{\mathrm{N}}(u)$ and $Q^{\mathrm{W}}(u)$ be the Nash and Wardrop aggregates under u (unique by Lemma 2). The *observation-only rule* of the companion paper, in its form without the market-power subsidy, posts

$$u^{t+1} = (1-\theta)u^t + \theta\,\kappa \odot Q(u^t), \qquad \theta \in (0,1],$$

using only the observed aggregate: the corrective charge at the observed deliveries. In the mean-field limit the market-power subsidy of the companion rule has no object, and this is the natural rule.

Lemma 10 (Responses are nonexpansive)**.** *For $X \in \{\mathrm{N},\mathrm{W}\}$ and charges u, u': $\|Q^X(u) - Q^X(u')\|_m^2 \le \langle u' - u, Q^X(u) - Q^X(u')\rangle$, hence $\|Q^X(u) - Q^X(u')\|_m \le \|u - u'\|_{m^{-1}}$.*

Proof. For $X = \mathrm{W}$ write the two optimality inequalities of Theorem 4(a) with the two equilibria $y = y^{\mathrm{W}}(u)$, $y' = y^{\mathrm{W}}(u')$: $\pi_i(y_i') - \langle a_i', MA + u\rangle \le \pi_i(y_i) - \langle a_i, MA + u\rangle$ and the symmetric one. Adding and summing over i gives $\|A - A'\|_m^2 + \langle A - A', u - u'\rangle \le 0$. For $X = \mathrm{N}$ the surcharges Ma_i, Ma_i' of Lemma 2(c) add $\sum_i \|a_i - a_i'\|_m^2 \ge 0$ to the left side. The second inequality follows by Cauchy–Schwarz in the dual pair $(\|\cdot\|_m, \|\cdot\|_{m^{-1}})$. □

Proposition 11 (Contraction and fixed point)**.** *Let $T_X(u) = \kappa \odot Q^X(u)$ for $X \in \{\mathrm{N},\mathrm{W}\}$.*

(a) $\|T_X(u) - T_X(u')\|_{\kappa^{-1}} \le \varrho\|u - u'\|_{\kappa^{-1}}$ with $\varrho = \max_k \kappa_k/m_k < 1$; the plain rule ($\theta = 1$) converges linearly with ratio ϱ.

(b) The unique fixed point is $u_X^* = \kappa \odot \tilde{Q}_X$, where $\tilde{Q}_X$ is the aggregate of any maximiser of $\tilde{\Phi}_X$, defined as Φ_X with $u = 0$ and the coupling $\frac{1}{2}\|A\|_m^2$ replaced by $\frac{1}{2}\|A\|_{m+\kappa}^2$ (the own terms of Φ_{N} keep the weight m).

(c) $\tilde{\Phi}_{\mathrm{W}} = \mathrm{SW}$, so $u_{\mathrm{W}}^* = \kappa \odot Q^{\mathrm{opt}}$ is the marginal-cost (Pigou) charge and the Wardrop equilibrium under u_{W}^* is socially optimal.

Proof. (a) $\|\kappa \odot \Delta Q\|^2_{\kappa^{-1}} = \sum_k \kappa_k \Delta Q_k^2 \le \varrho \|\Delta Q\|^2_m \le \varrho \|\Delta u\|^2_{m^{-1}} \le \varrho^2 \|\Delta u\|^2_{\kappa^{-1}}$, using $\kappa_k \le \varrho m_k$ twice and Lemma 10. (b) Let $\tilde{y}$ maximise $\tilde{\Phi}_X$ and $\tilde{Q}$ its aggregate. Lemma 1, applied to the convex term $\frac{1}{2}\|A\|^2_\kappa$, shows that $\tilde{y}$ maximises $\tilde{\Phi}_X + \frac{1}{2}\|A\|^2_\kappa - \langle \kappa \odot \tilde{Q}, A\rangle$, which is Φ_X with $u = \kappa \odot \tilde{Q}$. By Lemma 2(a), $\tilde{y}$ is an X-equilibrium under that charge, so $\kappa \odot \tilde{Q}$ is a fixed point; it is unique by (a). (c) $\frac{1}{2}\|A\|^2_{m+\kappa} = \frac{1}{2}\|Q\|^2_{b+2\kappa}$ is the quadratic part of (2). □

Part (a) is the contraction theorem of the companion paper ("the response is a proximal map"), restated for the time-expanded game and for both behaviours; part (c) is the classical marginal-cost toll (Beckmann et al., 1956), here reached by observation alone. Part (b) is what makes the finite-N analysis exact:

Theorem 12 (Charge convergence)**.** *Let $\tilde{Q}_{\mathrm{N}}, \tilde{Q}_{\mathrm{W}}$ be the Nash and Wardrop aggregates of the auxiliary game of Proposition 11(b) (coupling $m+\kappa$, own weight m). Then*

$$u^*_{\mathrm{N}} - u^*_{\mathrm{W}} = \kappa \odot (\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}), \qquad \|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}} = \|\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}\|_\kappa,$$

and consequently

- *(a) in general, $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}} \le \frac{1}{2}\tilde{\varrho}^{1/2}\big(\sum_i \|\tilde{a}^{\mathrm{W}}_i\|^2_m\big)^{1/2} \le \frac{1}{2}\big(\tilde{\varrho}\,\bar{s}\,\langle m, \tilde{Q}_{\mathrm{W}}\rangle\big)^{1/2}$ with $\tilde{\varrho} = \max_k \kappa_k/(m_k+\kappa_k)$;*
- *(b) for zone-separable carriers, $0 \le u^*_{\mathrm{W},k} - u^*_{\mathrm{N},k} \le \kappa_k \max_i \tilde{q}^{\mathrm{N}}_{ik} \le \kappa_k \bar{s}$ for every k;*
- *(c) for replicas (polyhedral types, linear profits), $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\| = O(1/n)$ with an instance-dependent constant.*

Moreover, for any charge-to-charge perturbation: if T is a ϱ-contraction with fixed point u^* and u' is a fixed point of a map T' with $\|T'u' - Tu'\| \le \varepsilon$, then $\|u' - u^*\| \le \varepsilon/(1-\varrho)$; with $T = T_{\mathrm{W}}$, $T' = T_{\mathrm{N}}$ this gives the form $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}} \le \|Q^{\mathrm{N}}(u^*_{\mathrm{N}}) - Q^{\mathrm{W}}(u^*_{\mathrm{N}})\|_\kappa/(1-\varrho)$.

Proof. The identity is Proposition 11(b). The auxiliary game is a game of the form of Section 3 with coupling $m' = m+\kappa$ except that each carrier's own term carries m instead of m'; Lemma 2(c) then gives own surcharges Ma_i instead of $M'a_i$, and the proofs of Theorems 4, 6 and 8 go through verbatim: in Theorem 4(a) one obtains $\|\Delta\tilde{Q}\|^2_{m'} \le \sum_i \langle \tilde{a}^{\mathrm{N}}_i, M(\tilde{a}^{\mathrm{W}}_i - \tilde{a}^{\mathrm{N}}_i)\rangle \le \frac{1}{4}\sum_i \|\tilde{a}^{\mathrm{W}}_i\|^2_m$, and $\|x\|^2_\kappa \le \tilde{\varrho}\|x\|^2_{m'}$ gives (a); in Theorem 6 the surcharge $m_k q_i \le m'_k q_i$ only strengthens (ii), so $0 \le \tilde{Q}_{\mathrm{W},k} - \tilde{Q}_{\mathrm{N},k} \le \max_i \tilde{q}^{\mathrm{N}}_{ik}$, which is (b) after multiplying by κ_k; (c) is Theorem 8. The perturbation statement: $\|u' - u^*\| = \|T'u' - Tu^*\| \le \|T'u' - Tu'\| + \|Tu' - Tu^*\| \le \varepsilon + \varrho\|u' - u^*\|$. □

So the finite-market fixed point lies *below* the Pigou charge in every coordinate when carriers are zone-separable, and the shortfall is at most one carrier's deliveries times the congestion slope. The rate of convergence of the charge is the rate of convergence of the aggregate in the auxiliary game; the factor $1/(1-\varrho)$ of the perturbation argument is not needed.

Theorem 13 (Welfare)**.** *Let y^{opt} be a social optimum with aggregate Q^{opt} and $\mathrm{SW}^{\mathrm{opt}} = \mathrm{SW}(y^{\mathrm{opt}})$.*

- *(a) For every charge v and every Wardrop equilibrium y under v with aggregate Q,*

$$\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \le \langle v - \kappa \odot Q^{\mathrm{opt}}, Q^{\mathrm{opt}} - Q\rangle - \tfrac{1}{2}\|Q - Q^{\mathrm{opt}}\|^2_b \le \tfrac{1}{2}\|v - \kappa \odot Q^{\mathrm{opt}}\|^2_{b^{-1}}.$$

- *(b) (Uniform $O(\bar{s})$ efficiency of the observation-only charge.) The Nash equilibrium under the finite-market fixed point u^*_{N} satisfies*

$$0 \le \mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u^*_{\mathrm{N}})) \le \tfrac{1}{2}\sum_i \|a^{\mathrm{opt}}_i\|^2_m - \tfrac{1}{2}\sum_i \|a^{\mathrm{N}}_i\|^2_m \le \tfrac{1}{2}\bar{s}\,\langle m, Q^{\mathrm{opt}}\rangle.$$

- *(c) For every charge u, $\big|\mathrm{SW}(y^{\mathrm{N}}(u)) - \mathrm{SW}(y^{\mathrm{W}}(u))\big| \le \frac{1}{2}\sum_i \|a^{\mathrm{W}}_i(u)\|^2_m + \|Q^{\mathrm{N}}(u) - Q^{\mathrm{W}}(u)\|_\kappa\,\|\bar{Q}(u) - \kappa^{-1} \odot u\|_\kappa$, where $\bar{Q}(u)$ is the midpoint of the two aggregates.*

Proof. Write $P^v(y) = \sum_i \pi_i - \frac{1}{2}\|Q\|_m^2 - \langle v, Q\rangle$, so $\mathrm{SW} = P^v + \langle v, Q\rangle - \frac{1}{2}\|Q\|_\kappa^2$. (a) y maximises P^v, which is the sum of a concave function and $-\frac{1}{2}\|Q\|_m^2$; by Lemma 1, y also maximises $\sum_i \pi_i - \langle v, Q\rangle - \langle MQ, \cdot\rangle$ evaluated at the aggregate, and the exact expansion of the quadratic term then gives $P^v(y) - P^v(y^{\mathrm{opt}}) \geq \frac{1}{2}\|Q - Q^{\mathrm{opt}}\|_m^2$. Hence $\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \leq -\frac{1}{2}\|\Delta Q\|_m^2 + \langle v, Q^{\mathrm{opt}} - Q\rangle - \frac{1}{2}\|Q^{\mathrm{opt}}\|_\kappa^2 + \frac{1}{2}\|Q\|_\kappa^2$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and $\frac{1}{2}\|Q\|_\kappa^2 - \frac{1}{2}\|Q^{\mathrm{opt}}\|_\kappa^2 = \langle \kappa \odot Q^{\mathrm{opt}}, \Delta Q\rangle + \frac{1}{2}\|\Delta Q\|_\kappa^2$; with $m - \kappa = b$ this is the first inequality, and maximising over ΔQ gives the second. (b) By Proposition 11(b), $y^{\mathrm{N}}(u_{\mathrm{N}}^*)$ maximises $\tilde{\Phi}_{\mathrm{N}} = \mathrm{SW} - \frac{1}{2}\sum_i \|a_i\|_m^2$; compare with y^{opt}. The last inequality uses $0 \leq a_{ik}^{\mathrm{opt}} \leq \bar{s}$. (c) $\mathrm{SW} = P^u - \frac{1}{2}\|Q - \kappa^{-1} \odot u\|_\kappa^2 + \frac{1}{2}\|u\|_{\kappa^{-1}}^2$. Lemma 2 gives $0 \leq P^u(y^{\mathrm{W}}) - P^u(y^{\mathrm{N}}) \leq \frac{1}{2}\sum_i \|a_i^{\mathrm{W}}\|_m^2 - \frac{1}{2}\sum_i \|a_i^{\mathrm{N}}\|_m^2$ (because y^{N} maximises $P^u - \frac{1}{2}\sum \|a_i\|_m^2$), and the difference of the two quadratic terms is $\langle Q^{\mathrm{N}} - Q^{\mathrm{W}}, \bar{Q} - \kappa^{-1} \odot u\rangle_\kappa$. □

Part (b) is uniform: unlike the aggregates (Proposition 5), the welfare of the observation-only charge converges at rate $O(\bar{s})$ with constant $\frac{1}{2}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance. The reason is that welfare is second order in the aggregate error near the optimum. Part (b) does not say that the charge helps when N is small: with few carriers the Cournot shortfall dominates and a charge that only adds the congestion cost can lower welfare (Section 10). For a fixed small N the rule of the companion paper with its market-power subsidy is the appropriate one.

Corollary 14 (Price of anarchy and leader gain)**.** *Let* $\mathrm{PoA}_X = \mathrm{SW}^{\mathrm{opt}}/\mathrm{SW}(y^X(0))$ *(welfare positive) and* $G_X = \mathrm{SW}(y^X(u_X^*)) - \mathrm{SW}(y^X(0))$ *the gain of the observation-only charge. Then* $|\mathrm{PoA}_{\mathrm{N}} - \mathrm{PoA}_{\mathrm{W}}| \leq \mathrm{PoA}_{\mathrm{N}}\mathrm{PoA}_{\mathrm{W}}\,|\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|/\mathrm{SW}^{\mathrm{opt}}$ *and* $|G_{\mathrm{N}} - G_{\mathrm{W}}| \leq [\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u_{\mathrm{N}}^*))] + |\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|$*; by Theorems 13 and 4–8 both tend to zero at rate* $O(\bar{s}) + O(\|Q^{\mathrm{N}}(0) - Q^{\mathrm{W}}(0)\|_\kappa)$*, that is* $O(\bar{s})$ *for separable carriers and replicas and* $O(\bar{s}^{1/2})$ *in the worst case.*

Proof. Direct from the definitions, $\mathrm{SW}(y^{\mathrm{W}}(u_{\mathrm{W}}^*)) = \mathrm{SW}^{\mathrm{opt}}$ (Proposition 11(c)) and Theorem 13(b),(c) at $u = 0$. □

6. Cautious carriers in the limit

Only two pieces of the imprecise-probability layer of the companion paper are used: the Γ-maximin surcharge and the scaling of the belief width needed for credal absorption. Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \geq 0$; a carrier believes that the mean rival aggregate lies in the box $\{A' : |A'_k - A_{-i,k}| \leq \delta w_k\}$.

Proposition 15 (Γ-maximin carriers)**.** *(a) (companion paper, Lemma "Γ-maximin surcharge".) Because deliveries are nonnegative and the coupling is affine, the lower expectation of every plan is attained at the upper corner* $A_{-i} + \delta w$*. Hence the Γ-maximin equilibria, finite or mean-field, are the precise equilibria under the charge* $u + \delta\, m \odot w$*:* $Q^{X,\delta}(u) = Q^X(u + \delta m \odot w)$*.*

(b) $\|Q^{X,\delta}(u) - Q^X(u)\|_m \leq \delta\|w\|_m$*; for zone-separable carriers,* $0 \leq Q_k^X(u) - Q_k^{X,\delta}(u) \leq \delta w_k$ *for* $X = \mathrm{W}$*.*

(c) The observation-only rule under Γ-maximin carriers, $u \mapsto \kappa \odot Q^{X,\delta}(u)$*, contracts with the same ratio* ϱ *(it needs no knowledge of* δ*); its fixed point* u_δ^* *satisfies* $\|u_\delta^* - u_X^*\|_{\kappa^{-1}} \leq \varrho^{1/2}\delta\|w\|_m/(1 - \varrho)$*.*

(d) In the mean-field game, the equilibrium under u_δ^* *loses at most* $\frac{1}{2}\delta^2\|m \odot w\|^2_{(m+\kappa)^{-1}}$ *of welfare. A city that knows* δw *and posts* $u_\delta^* - \delta m \odot w$ *recovers the optimum.*

Proof. (a) is the lemma of the companion paper; in the time-expanded model the coordinates are zone-hours, and nothing else changes. (b) Lemma 10 with $u' = u + \delta m \odot w$: $\|\delta m \odot w\|_{m^{-1}} = \delta\|w\|_m$. In the separable case the proof of Theorem 6 with a common price increase $\delta m_k w_k$ in coordinate k (in place of the private surcharges) gives the coordinatewise statement: the Wardrop aggregate falls, and by at most the price increase divided by m_k. (c) The map is

$T_X(\cdot + \delta m \odot w)$, a ϱ-contraction in $\|\cdot\|_{\kappa^{-1}}$ (Proposition 11). Its distance to T_X at any u is $\|Q^X(u + \delta m \odot w) - Q^X(u)\|_\kappa \le \varrho^{1/2}\|\cdot\|_m \le \varrho^{1/2}\delta\|w\|_m$; apply the perturbation statement of Theorem 12. (d) The carriers face $v = u^*_\delta + \delta m \odot w$ and $u^*_\delta = \kappa \odot Q$ at the fixed point, so in Theorem 13(a) $v - \kappa \odot Q^{\mathrm{opt}} = \kappa \odot \Delta Q + \delta m \odot w$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and the bound becomes $-\langle \delta m \odot w, \Delta Q\rangle - \frac{1}{2}\|\Delta Q\|^2_{b+2\kappa} \le \frac{1}{2}\delta^2\|m \odot w\|^2_{(b+2\kappa)^{-1}}$, with $b + 2\kappa = m + \kappa$. □

So cautious carriers behave like carriers facing a tax of $\delta m_k w_k$ per delivery; the observation-only city cannot tell this tax apart from cost, and over-charges by about that amount. The welfare cost is second order in δ and of Harberger form.

Corollary 16 (Belief width needed for absorption)**.** *Let $\sigma_{\max}$ be the largest range of one carrier's deliveries at one coordinate in units of w. The uniform credal-absorption bound of the companion paper, $\delta_0^E \le \sigma_{\max} \min\{N-1, d+1, 1+\sqrt{d\ln(2d)/2}\}$, is at most $(d+1)\bar{s}/\min_k w_k$. For n-splits of a fixed instance $\bar{s} \propto 1/n$, so the width at which a rounded pure profile becomes an E-admissible equilibrium is $O(d/n)$ and vanishes in the limit; the N-free terms bind exactly when $N > d+1$.*

Proof. $\sigma_{\max} \le \bar{s}/\min_k w_k$ and $\min\{\cdot\} \le d+1$. □

With $d = ZH$ coordinates the terms $d+1$ and $1+\sqrt{d\ln(2d)/2}$ become relevant only for very many carriers (for Cairo with 179 active zone-hours in the experiments, $N > 180$), which is the regime of gig couriers and small operators. The corollary is algebra on a theorem of the companion paper; its content is the regime statement.

Proposition 17 (Time-of-day charges can move the worst case of travel times)**.** *In the routing layer of the companion paper the cost of a route is nondecreasing in every arc travel time, and the upper prevision over a probability box with unknown dependence is attained at the upper (slowest) marginals. If instead a period-dependent charge is paid at the arrival time, monotonicity can fail: with one arc whose travel time T has a p-box containing all laws on $[0,2]$, and cost $T + U \cdot \mathbf{1}\{T < 1\}$ with $U > 2$, the upper prevision equals $\sup_{T\in[0,2]}(T + U\mathbf{1}\{T<1\}) = 1 + U$ (approached as $T \uparrow 1$), whereas the slowest law ($T = 2$) costs 2.*

Proof. The p-box contains every Dirac law on $[0,2]$, so the upper prevision is the supremum of the cost; on $[0,1)$ the cost is $T + U < 1 + U$ and tends to $1+U$, on $[1,2]$ it is at most $2 < 1 + U$. □

When charges increase over the day (or are constant), the arrival-time cost is again non-decreasing in travel time and the corner property of the companion paper is restored; with decreasing charges, the robust route cost must be computed over the whole box. In the experiments below travel times are deterministic, so this proposition limits the transfer of the routing layer of the companion paper to the time-expanded model rather than affecting the computations.

7. Vehicle-count congestion

Under D2 the congestion variable counts vehicles. Let mean-field types $\theta \in [\Theta]$ choose exposures $\tau_\theta \in \mathcal{T}_\theta \subset \mathbb{R}^Z_{\ge 0}$ (vehicle-hours per zone; compact convex) with concave value V_θ, and let a type-θ vehicle dwell $h_{\theta z} > 0$ hours in zone z, so that it contributes $\tau_{\theta z}/h_{\theta z}$ vehicles. The congestion variable is $X_z = \sum_\theta \tau_{\theta z}/h_{\theta z}$ and a vehicle-hour in z costs $g_z X_z$, $g_z > 0$. A mean-field equilibrium is a profile in which each τ_θ maximises $V_\theta(\tau) - \langle \tau, g \odot X\rangle$ at the realised X. The coupling operator $F_\theta(\tau) = g \odot X(\tau)$ has Jacobian blocks $\partial F_\theta/\partial \tau_{\theta'} = \mathrm{diag}_z(g_z/h_{\theta' z})$.

Proposition 18 (Existence and the rank-one condition)**.** *(a) An equilibrium exists.*

(b) If $h_{\theta z} = \eta_\theta \zeta_z$ (rank one), the game is a weighted potential game with weights $1/\eta_\theta$ and concave potential $\Psi(\tau) = \sum_\theta \eta_\theta^{-1} V_\theta(\tau_\theta) - \frac{1}{2}\sum_z g_z \zeta_z X_z^2$; equilibria are the maximisers of Ψ, and X is unique when the V_θ are concave.

(c) *Positive weights w_θ with $w_\theta\,\partial F_\theta/\partial\tau_{\theta'}$ symmetric for all θ,θ' exist iff $h_{\theta z}/h_{\theta' z}$ does not depend on z, i.e. iff h is rank one.*

(d) *With two types in one zone and $h_1 \neq h_2$, the symmetric part of the coupling matrix $g\left(\begin{smallmatrix} 1/h_1 & 1/h_2 \\ 1/h_1 & 1/h_2 \end{smallmatrix}\right)$ has determinant $-g^2(1/h_1 - 1/h_2)^2/4 < 0$, so the unweighted operator is not monotone.*

Proof. (a) Each type's best-response correspondence at a given X is nonempty, convex-valued and upper hemicontinuous (Berge), and X is continuous in τ; Kakutani's theorem on $\prod_\theta \mathcal{T}_\theta$ gives a fixed point. (b) $\partial\Psi/\partial\tau_{\theta z}$ of the coupling part is $-g_z\zeta_z X_z/(\eta_\theta\zeta_z) = -\eta_\theta^{-1} g_z X_z$, which is η_θ^{-1} times the marginal coupling cost of type θ; Ψ is concave and the equivalence of equilibria and maximisers follows as in Lemma 2(a); X is a linear image on which Ψ is strictly concave. (c) Symmetry means $w_\theta g_z/h_{\theta' z} = w_{\theta'} g_z/h_{\theta z}$, i.e. $h_{\theta z}/h_{\theta' z} = w_{\theta'}/w_\theta$ for all z. (d) Direct computation. □

In the finite game, deciding whether a pure equilibrium exists under D2 is strongly NP-complete (companion paper, Theorem "NP-completeness under D2"); in the convexified mean-field game an equilibrium always exists, so the limit removes the existence problem. What the limit does not remove is non-monotonicity: with dwell times that are not rank one there is no weighting under which the game is a potential game.

Conjecture 19. *For rank$(h) \geq 2$ there are instances with several isolated mean-field equilibria, and computing one is PPAD-hard in the number of types.*

Neither part has been checked; no reduction has been designed. The rank-one condition is plausible for a fleet of one vehicle class (dwell time proportional to a zone factor) and implausible for mixed fleets of vans and cargo bikes.

8. Computation

Equilibria as quadratic programs. By Lemma 2, every object of the paper maximises a concave quadratic potential over a product of fleet polytopes: Nash (Φ_{N}), Wardrop (Φ_{W}), the social optimum (Φ_{W} with coupling $m+\kappa$) and the two observation-only fixed points (Proposition 11(b)). We introduce the deliveries a_i and the aggregate Q as variables, so that the Hessian is diagonal, and solve the program with the interior-point method Clarabel (Goulart and Chen, 2024) at relative gap 10^{-13}. The accuracy of each solution is measured by the Frank–Wolfe (duality) gap Γ of the potential, and $\|Q - Q^*\|_m \leq (2\Gamma)^{1/2}$ because the potential is 1-strongly concave in Q in the m-norm; every reported gap carries this error bound, which is at most 0.135 of the gap in the replica runs.

Replicas cost the same as one carrier per type. By Lemma 3 the Nash equilibrium of n replicas per segment is one program in the segment totals with own weight $1/n$; the experiments therefore reach $N = 1536$ at the cost of $N = 6$. Heterogeneous splits need one block per carrier.

Learning dynamics. The conditional-gradient (Frank–Wolfe) method with exact line search is the natural decentralised dynamics: in each round every carrier computes one best response to the current zone-hour prices (one oracle call: the tour with the largest profit per vehicle-hour) and the profile moves toward it. For potential MFGs this is fictitious play in its conditional-gradient form (Cardaliaguet and Hadikhanloo, 2017; Lavigne and Pfeiffer, 2023); its gap decreases at rate $O(1/k)$. It is included as the method a decentralised market would follow and as an independent check of the interior-point solutions (tests in `mfg/tests`).

Best responses on the time-expanded graph. A carrier's best response to prices is a longest-path problem on the acyclic graph G_H with a constraint that a zone is served at most once per tour. With at most three stops per tour, as here, the tours are enumerated (one ordering per stop set, the one with least travel time, for each start hour); with longer tours the best response is a prize-collecting elementary path problem, NP-hard in general, and the labelling algorithms of the companion paper apply.

9. Data and stylised hourly profile

Cities. We use the nine processed cities of the companion paper (zones, populations, zone-to-zone travel times on the major road graph, depot scores and the parameter table with grades A/B/C; Table B.1). For each city and seed we draw six *segments* of small carriers. A segment has a depot (high depot score, at least 3 km apart), eight customer zones drawn around the depot by population times a gravity decay, lots per stop as in the calibration of the companion paper, and a fleet budget in vehicle-hours equal to the segment's daily parcels divided by drops per vehicle-day times the shift length. The segment's demand share is 1/6 of the zone's parcels. Tours start in any of the ten hours, visit one to three customer zones, deliver one lot per stop, and end at the depot by 18:00; this gives between 1476 and 3095 tours and between 176 and 359 active zone-hours per instance.

Calibration. As in the companion paper: $\alpha_k = 1.5\times$ delivery fee, $b_k = \alpha_k/(2Q_k^{\text{full}})$ with Q_k^{full} the slice's demand in the hour, in-zone hours per delivery $r = 0.6\times$ shift/drops, and $\kappa_k = \gamma c_k r^2$ with c_k the derivative of the BPR curve at the hourly congestion level, scaled by the slice multiplicity. A tour costs the hourly vehicle cost (wage plus fuel) times its duration plus one hour.

The hourly profile is an assumption. The congestion level $\text{cong}(h)$ is the TomTom all-day value with two Gaussian rush-hour bumps (width one hour) centred at 08:30 and 17:30 whose heights reproduce the TomTom morning and evening peak values of each city (anchors graded A; the shape graded C; Moscow's peaks, absent from the index, imputed from the other cities' ratios, graded C). The demand share $\psi(h)$ has a late-afternoon bump of +60% and a late-morning bump of +30%, normalised to mean one (graded C; no data). An amplitude parameter scales all deviations from a flat profile (amplitude zero is the static model split into ten equal hours). Results that depend on the profile are reported as sensitivity analyses over the amplitudes 0, 0.5, 1 and 1.5, and nothing in the paper relies on the profile being realistic.

10. Computational study

Protocol. Main study: nine cities, five seeds, profile amplitude 1, replica splits $k \in \{1, 2, 4, \ldots, 256\}$ per segment ($N = 6k$ carriers, $N \le 1536$), 405 instances. Heterogeneous splits (Dirichlet(2) fleet shares and individual routing-cost multipliers in $[0.85, 1.15]$): three seeds, $k \le 16$, 135 instances, each compared with the Wardrop equilibrium of the same population. Sensitivity: amplitudes 0, 0.5, 1.5 (three seeds) and a congestion-heavy variant with $\kappa \times 10$ (three seeds, 243 instances). All runs are resumable scripts in `mfg/experiments`; results are in `mfg/results`. Confidence intervals are t-intervals over seeds of per-seed least-squares slopes.

E1: the gap behaves like one carrier's delivery. Figure 1 shows the relative gap $\|Q^{\text{N}} - Q^{\text{W}}\|_m / \|Q^{\text{W}}\|_m$ against N. Over all cities and seeds the slope for $N \ge 24$ is -0.88 $[-0.90, -0.86]$, between -0.93 and -0.77 by city. The slope is not -1 because the share itself does not fall exactly like $1/N$: the largest delivery of one carrier, s_N, scales like $N^{-0.92}$ (segments deliver more in total as the market becomes more competitive), and the gap against s_N has slope 0.95 $[0.92, 0.97]$. In units of s_N the gap stays near one: median 0.98, range 0.35–1.86 (right panel). The general bound of Theorem 4 held in 405/405 instances (largest ratio 0.99). For heterogeneous splits the slope over $N = 12$ to 96 is -0.79 $[-0.83, -0.75]$, the ratio to s_N has median 0.76 and maximum 1.57,

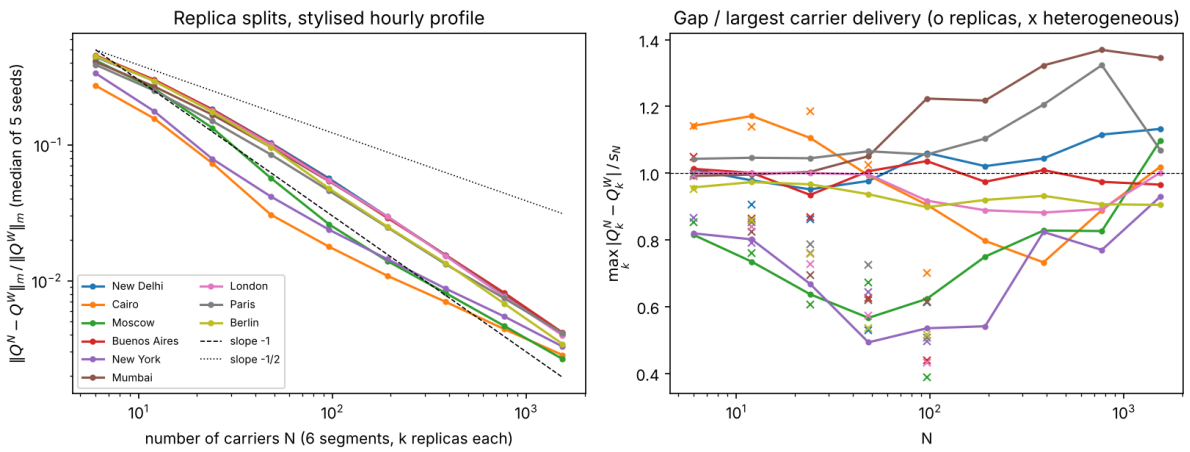


Figure 1: Left: relative aggregate gap against the number of carriers (replicas, median of five seeds). Right: largest coordinate gap in units of the largest carrier delivery s_N (circles: replicas; crosses: heterogeneous splits).

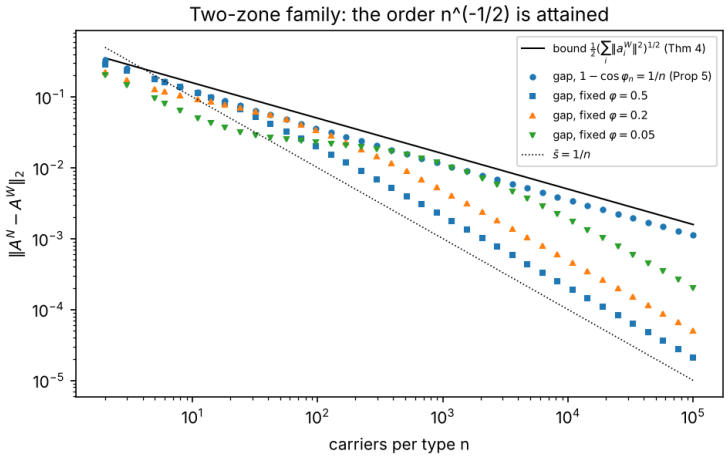


Figure 2: The two-zone family of Proposition 5. With $1-\cos\varphi_n = 1/n$ the gap follows the bound of Theorem 4 at ratio $1/\sqrt{2}$; for a fixed angle it eventually falls like $1/n$, with a constant $1/\sin\varphi$.

and the general bound held in 135/135. The heterogeneous slope is flatter and its range of N is short; we read it as consistent with $O(s_N)$ but not as evidence of a rate. No city comes close to the adversarial behaviour of Proposition 5: the theory says that such behaviour needs nearly collinear segment types with different own surcharges, and the city instances do not produce it.

E2: the charge and the welfare. The calibrated congestion is weak relative to demand: ϱ lies between 0.009 and 0.517, the mean-field price of anarchy is 0.026% of optimal welfare (median; maximum 0.591%), and the welfare cost of the finite market is almost entirely Cournot withholding: PoA-1 is 21.4% at $N = 6$ (range 8.7–28.1%) and 0.020% at $N = 1536$ (Figure 3). The observation-only fixed point approaches the Pigou charge with slope -0.91 $[-0.92, -0.89]$ (calibrated) and -0.90 $[-0.92, -0.89]$ (congestion $\times 10$, ϱ up to 0.914). The welfare bound of Theorem 13(b) held in 405/405 instances; the realised loss was at most 0.272 of the bound.

The negative result is the leader gain. With few carriers the observation-only charge *lowers* welfare: median -0.45% at $N = 6$ (calibrated) and -3.00% with congestion $\times 10$, because it taxes deliveries that Cournot carriers already withhold. With congestion $\times 10$ the gain is positive in every city and seed from $N = 192$ on and reaches 2.28% at $N = 1536$ against 2.36% in the limit. With the calibrated congestion the attainable gain is tiny (0.026% in the limit) and the charge never helps in all instances at once. The flat rule of the companion paper, which adds

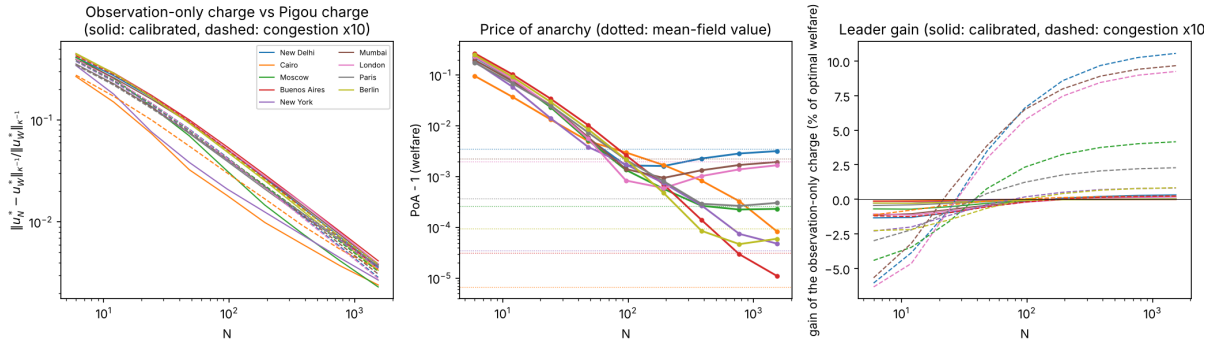


Figure 3: Left: distance of the finite-market observation-only charge from the Pigou charge. Middle: price of anarchy at N carriers (dotted: limit). Right: welfare gain of the observation-only charge (solid: calibrated congestion; dashed: congestion $\times 10$).

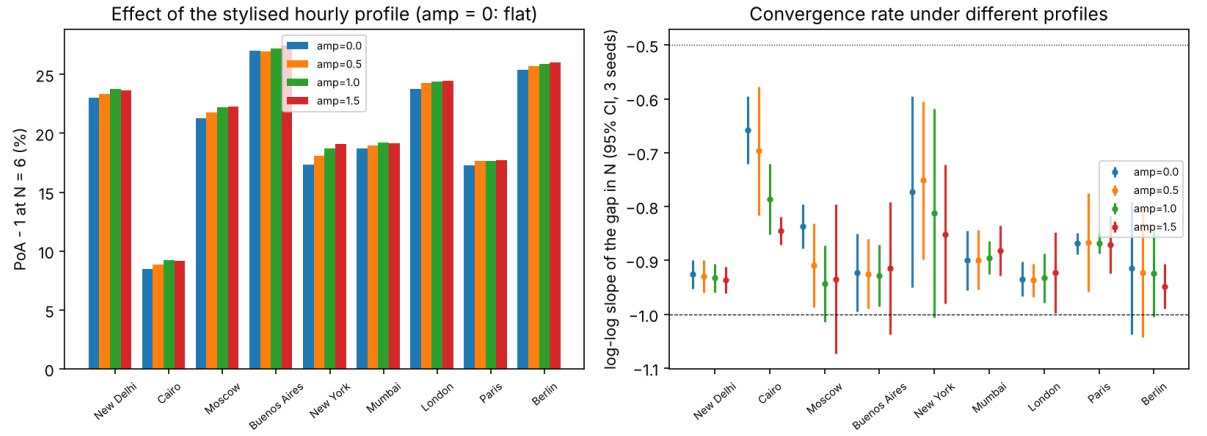


Figure 4: Sensitivity to the amplitude of the stylised hourly profile (three seeds). Left: PoA-1 at $N = 6$. Right: gap slopes with 95% intervals.

the market-power subsidy, is the right rule for few carriers (E5).

E3: the hourly profile matters little for these questions. With amplitudes 0, 0.5, 1 and 1.5 the median PoA-1 at $N = 6$ is 20.7, 21.4, 21.6 and 21.8%, and the pooled gap slopes are -0.86, -0.87, -0.89 and -0.90 (Figure 4). The convergence results do not depend on the profile, as the theory predicts (the bounds are uniform in the data), and the economic quantities move by about one percentage point. This is a weak result in the sense that it says little about real time-of-day effects: with a stylised profile, it shows only that our conclusions are not artefacts of a particular shape.

E4: cautious carriers and the belief width. With the belief box in units of the median lot, the mean-field aggregate moves by at most 0.973 of the bound $\delta\|w\|_m$ of Proposition 15(b) (median 0.614). The observation-only charge under cautious carriers loses 1.17% of optimal welfare at $\delta = 1$ and 11.85% at $\delta = 4$ (median), within the bound of Proposition 15(d) in 675/675 cases (median ratio 0.459); the charge shift satisfied part (c) in 675/675 cases (Figure 5). The absorption width of Corollary 16, relative to the mean demand of a zone-hour, falls from 481.21 at $N = 6$ to 0.698 at $N = 24576$ (Figure D.11 in the appendix); with d between 179 and 359 active zone-hours the N-free terms bind only beyond a few hundred carriers. This is a weak result: the uniform bound uses the largest delivery a carrier *could* make at one zone-hour (its whole fleet there), and even with twenty-four thousand carriers it remains of the order of a zone-hour's demand, so the absorption theorem does not give a practically small belief width on

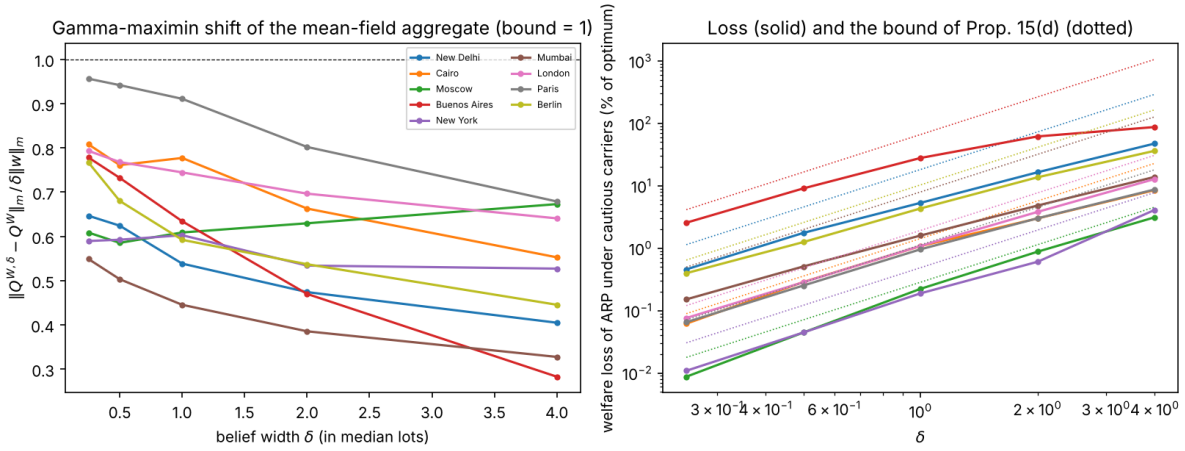


Figure 5: Γ-maximin carriers ($N = 1536$). Left: shift of the mean-field aggregate relative to the bound. Right: welfare loss of the observation-only charge and the bound $\frac{1}{2}\delta^2\|m \odot w\|^2_{(m+\kappa)^{-1}}$.

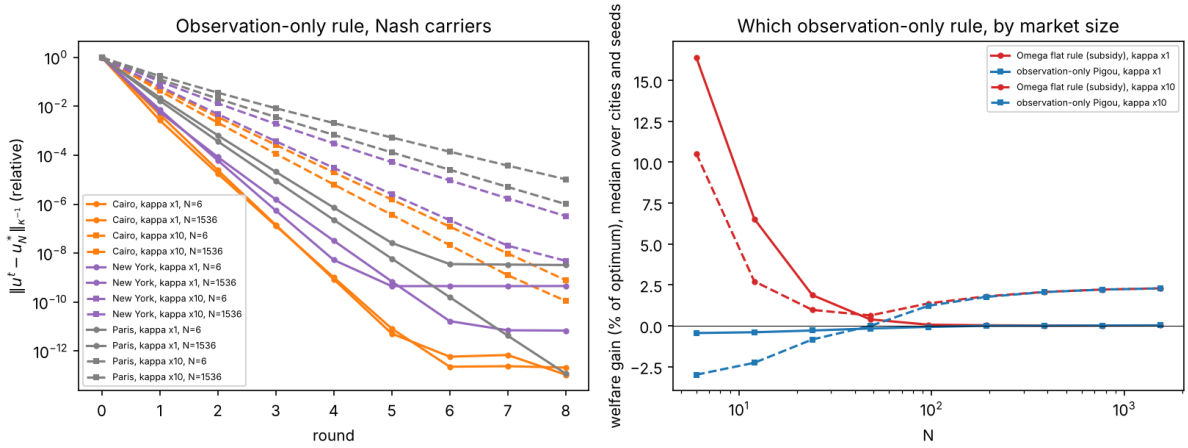


Figure 6: Left: plain observation-only iteration (Nash carriers), distance to the fixed point. Right: welfare gain of the flat rule of the companion paper and of the observation-only Pigou rule against N (median over cities and seeds).

these instances.

E5: which observation-only rule. Figure 6 (right) compares the observation-only Pigou rule with the flat rule of the companion paper, $u \leftarrow \kappa Q - m \odot \bar{d}$, where $\bar{d}$ is the observed mean delivery per active carrier (damped iteration). At $N = 6$ the flat rule gains 16.41% (calibrated; 10.52% with congestion $\times 10$) where the Pigou rule loses -0.45% (-3.00%); at $N = 1536$ the flat rule gains 0.019% (2.284%) and its charge is within 0.025 (relative, congestion $\times 10$) of the Pigou charge. The subsidy vanishes like the share, so the two rules meet in the limit. The plain rule contracted in every run at a per-round ratio of at most 0.945 times ϱ, counting the rounds before the distance reached 10^{-6} of its initial value, below which solver accuracy dominates (left panel).

E6: cost. The replica reduction makes the Nash equilibrium for $N = 1536$ as cheap as for $N = 6$ (well under a second per program). Heterogeneous splits grow with N: at $N = 96$ one interior-point solve took up to 90 seconds (Wardrop, whose solution set in the individual variables is not a singleton) and about 20 seconds (Nash). Plain Frank–Wolfe is slow: within 3000 rounds it reached a relative duality gap of 10^{-4} in 0/36 runs, and its relative gap after 3000 rounds was between 0.29% and 1.27% of the potential (Figure D.12 in the appendix). It is adequate as a model of how a decentralised market adjusts, not as a solver for the precision that the rate

study needs; all reported equilibria come from the interior-point method.

11. Who gains and who loses

The welfare results above add up three groups: carriers, buyers of deliveries and the city. This section separates them. The accounting is that of Section 3: in zone-hour k the price is $\alpha_k - b_k Q_k$, a delivery carries the delay cost $\kappa_k Q_k$, and the city posts a charge u_k per delivery. For a profile with aggregate Q and carrier profit vector π,

$$\text{consumer surplus } \mathrm{CS} = \tfrac{1}{2}\langle b, Q^2\rangle, \qquad \text{congestion cost } \mathrm{CC} = \langle \kappa, Q^2\rangle,$$
$$\text{charge revenue } R = \langle u, Q\rangle, \qquad \text{carrier profit } \Pi = \sum_i \pi_i - \langle b, Q^2\rangle - \mathrm{CC} - R,$$

and $\mathrm{SW} = \mathrm{CS} + \Pi + R$ exactly (unit test in `mfg/tests/test_stake.py`; the identity holds to 6.6e-16 relative error in all 405 instances). Charges are a transfer: they move welfare between Π and R and change SW only through the deliveries. In this model the delay cost is borne by the delivery fleet, so residents carry no monetary congestion cost; their stake is the demand side (CS) and, in Section 11.3, the vehicle-hours spent in their zone.

Three rules. All rules are evaluated at the Nash equilibrium of the finite market under the posted charge, on the instances of Section 10 (nine cities, five seeds, $N = 6k$, $k = 1, \dots, 256$). *No charge*: $u = 0$. *Observation-only*: the finite-market fixed point $u = \kappa \odot Q(u)$ of Proposition 11, the Pigou charge without any market-power correction. *Pigou minus market power* (the rule of the companion paper): start from $u = \kappa \odot Q^{\mathrm{opt}} - m \odot \bar{a}$, with $\bar{a}_k$ the delivery per active carrier at the social optimum, and correct for 25 damped rounds, $u \leftarrow u + 0.8\, m \odot (1 + 1/n_k) \odot (Q(u) - Q^{\mathrm{opt}})$ (n_k active carriers), until the Nash aggregate matches Q^{opt}. The median relative M-norm residual of the match is 0.0002 (maximum 0.013). The rule reproduces the optimal aggregate; with several carrier types it need not reproduce the optimal individual plans, so it can fall short of $\mathrm{SW}^{\mathrm{opt}}$ (median shortfall 1.08 % at $N = 6$ and below 0.01 % at $N = 1536$, against 17.66 % and 0.02 % without charge and 18.35 % for the observation-only rule at $N = 6$).

11.1. The charge that implements the optimum

For one segment split into N identical carriers the optimal charge has a closed form.

Proposition 20 (Pigou minus market power)**.** *Let the N carriers be identical replicas of one type (Lemma 3) and let Q^{opt} be the aggregate of a social optimum. The charge*

$$u_N^* = \kappa \odot Q^{\mathrm{opt}} - \tfrac{1}{N}\, m \odot Q^{\mathrm{opt}}$$

makes the Nash aggregate equal to Q^{opt} and its welfare equal to $\mathrm{SW}^{\mathrm{opt}}$, for any coupling of the plans across zone-hours. It is the Pigou charge $\kappa \odot Q^{\mathrm{opt}}$ minus a market-power term of exact size $\frac{1}{N} m_k Q_k^{\mathrm{opt}}$. Zone-hour k is taxed if $N > m_k/\kappa_k = 1/\varrho_k$ and subsidised if $N < 1/\varrho_k$. In the one-zone case with a slack fleet budget, $Q^{\mathrm{N}} = \frac{N}{N+1} Q^{\mathrm{W}}$ with $Q^{\mathrm{W}} = \alpha/m$ and $Q^{\mathrm{opt}} = \alpha/(m+\kappa)$, and $u_N^* = \alpha(\kappa - m/N)/(m+\kappa)$.

Proof. By Lemma 3 with one type, the type total Y of a Nash equilibrium maximises $\Phi(Y) = \pi(Y) - \langle u, A\rangle - \frac{1}{2}(1 + \frac{1}{N})\|A\|_m^2$, $A = LY$, over the convex fleet polytope, and each carrier delivers A/N. At $u = u_N^*$ the gradient of Φ at the social optimum Y^{opt} is $\nabla\pi - L^\top[(\kappa - \frac{m}{N}) \odot Q^{\mathrm{opt}} + (1 + \frac{1}{N}) m \odot Q^{\mathrm{opt}}] = \nabla\pi - L^\top[(m+\kappa) \odot Q^{\mathrm{opt}}]$, the gradient of the social objective $\pi - \frac{1}{2}\|A\|_{m+\kappa}^2$ at Y^{opt}. The variational inequality of Y^{opt} for the social problem is therefore that of Φ; Φ is concave, so Y^{opt} maximises it, and the Nash aggregate is unique (Lemma 2(b)). Welfare depends on the profile only through π and Q. One zone: the Nash first-order condition $\alpha - u = (1 + \frac{1}{N}) m Q$ gives $Q^{\mathrm{N}} = \frac{N}{N+1}\alpha/m$ at $u = 0$, and $u = \alpha - (1 + \frac{1}{N}) m \frac{\alpha}{m+\kappa} = \alpha \frac{\kappa - m/N}{m+\kappa}$ at $Q = Q^{\mathrm{opt}}$. □

Table 1: Change with respect to no charge, in % of welfare without charge (median of 45 instances). ΔΠ: carrier profit net of the charge; ΔCS: consumer surplus; ΔCC: congestion cost; R: charge revenue (negative: subsidy paid by the city); ΔSW: welfare; last column: R as a share of the welfare under the rule. Medians of each column are taken separately.

N	rule	ΔΠ	ΔCS	ΔCC	R	ΔSW	R/SW
6	observation-only	-1.1	-1.0	0.0	+1.6	-0.5	+1.6
	Pigou minus market power	+132.0	+69.1	+2.2	-181.9	+19.6	-151.9
24	observation-only	-0.7	-2.4	-0.1	+2.7	-0.3	+2.8
	Pigou minus market power	+13.4	+21.6	+0.7	-36.9	+1.9	-36.1
192	observation-only	-0.4	-1.8	-0.1	+3.3	0.0	+3.3
	Pigou minus market power	+0.3	+0.1	0.0	-1.9	0.0	-1.9
1536	observation-only	-0.1	-1.6	-0.1	+3.4	0.0	+3.4
	Pigou minus market power	-0.1	-1.1	-0.1	+3.0	0.0	+3.0

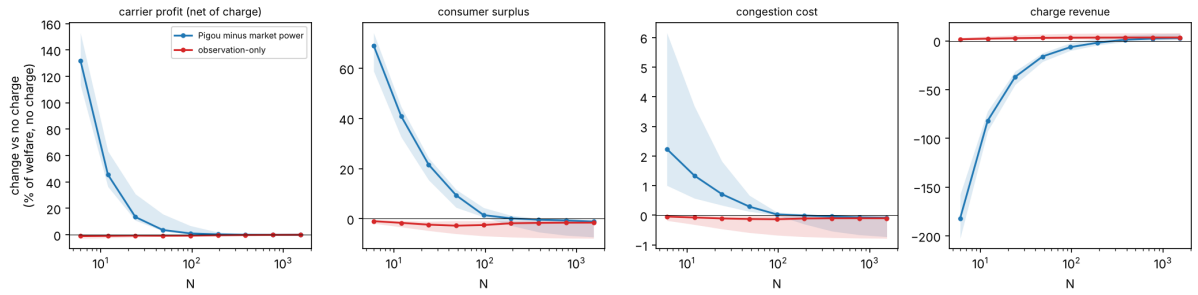


Figure 7: Change of each welfare term against no charge (% of welfare without charge; median and interquartile range over nine cities and five seeds). Blue: Pigou minus market power; red: observation-only.

The proposition holds in the numerical check on the nine cities with one segment per city (three seeds, $N = 1, 2, 3, 6, \ldots, 1536$, 324 instances): under u_N^* the Nash aggregate equals Q^{opt} to a relative M-norm error of at most 4.7e-09 and welfare to 5.7e-11. Without a charge a single-segment market loses 1.42 % of optimal welfare at $N = 6$ (median) and the pure Pigou charge loses 2.04 %; at $N = 1536$ the losses are 0.02 % and below 0.001 %. The charge u_N^* is a subsidy in every zone-hour at $N = 6$, in 9 % of them at $N = 192$ and in none at $N = 1536$, where it equals 0.95 times the Pigou charge in total. The one-zone relation $Q^{\mathrm{N}} = \frac{N}{N+1} Q^{\mathrm{W}}$ is exact only while the fleet budget is slack: it held to 10^{-9} in six of the nine cities and deviated by up to 16.5 % in the others, where the budget binds; the charge formula does not use it. With several carrier types u_N^* is not available, because the terms $m \odot a_\theta$ differ across types and one posted charge cannot cancel all of them; the rule of the previous paragraph is the numerical substitute.

11.2. The city council

Table 1 gives, for $N \in \{6, 24, 192, 1536\}$ and the two rules that charge, the change of each term relative to no charge, as a percentage of welfare without charge (medians over nine cities and five seeds; city values are in `results/summary_stake_who_city.csv`). Figure 7 shows all nine sizes.

Revenue and budget. With many carriers the council collects about 3.4 % of welfare under the observation-only rule and 3.0 % under the market-power rule at $N = 1536$, the Pigou revenue in the limit. With few carriers the picture reverses. At $N = 6$ the observation-only charge still raises 1.6 % but changes welfare by −0.5 %: it taxes deliveries that Cournot carriers already withhold. The market-power rule raises welfare by 19.6 % at a net cost to the council: its revenue is −181.9 % of the no-charge welfare (−151.9 % of the welfare under the rule), negative in 45/45 instances, and the charge is a subsidy in 98 % of the zone-hours. It removes market power by

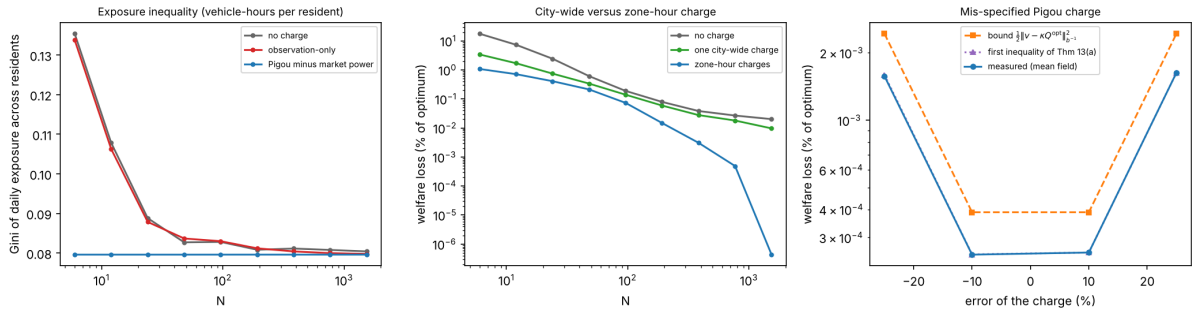


Figure 8: Left: Gini of daily exposure across residents of the served zones (median of 45 instances). Middle: welfare loss with no charge, with the best single city-wide charge and with zone-hour charges (Pigou minus market power). Right: mean-field welfare loss of a Pigou charge scaled by $1+\varepsilon$, with the bound of Theorem 13(a) (median over instances; the first inequality of the theorem is indistinguishable from the measured loss).

paying carriers to deliver more. At $N = 24$ the transfer is still $-36.9\,\%$ and at $N = 192$ it is $-1.9\,\%$. A council that cannot run a deficit would have to recover the transfer by a lump-sum fee or settle for a smaller correction; Section 12.1 examines the fee. Budget neutrality holds only in large markets.

One city-wide charge or zone-hour charges. If the council may post a single per-delivery charge $\bar{u}$ for the whole city, the best scalar (grid plus bounded search; `results/uniform.csv`) leaves a welfare loss of $3.41\,\%$ of the optimum at $N = 6$, against $1.08\,\%$ for zone-hour charges and $17.66\,\%$ with no charge; it captures $85\,\%$ of the gain of the zone-hour rule (median; range 74–93 %). The best scalar is a subsidy for small N and its median turns positive between $N = 192$ and $N = 384$. At $N = 1536$ the gain captured is $68\,\%$ (range 4–89 %), but the amounts are tiny ($0.01\,\%$ lost against $0.02\,\%$ without charge). In the mean-field limit the best single charge loses $0.006\,\%$ of the optimum (maximum $0.175\,\%$), zone-hour charges lose nothing, and no charge loses $0.026\,\%$: the best city-wide charge captures $78\,\%$ of the possible gain (range 33–90 %). Differentiation by zone-hour adds a modest share of the gain, and in absolute terms the welfare at stake is small with the calibrated congestion.

A wrong charge. Let the council post $(1+\varepsilon)$ times the Pigou charge in the mean-field market, $|\varepsilon| \in \{10\%, 25\%\}$. Theorem 13(a) bounds the loss by $\frac{1}{2}\|v - \kappa \odot Q^{\mathrm{opt}}\|^2_{b-1} = \frac{1}{2}\varepsilon^2\|\kappa \odot Q^{\mathrm{opt}}\|^2_{b-1}$. The measured median loss is $0.0003\,\%$ of the optimum for $|\varepsilon| = 10\%$ (maximum $0.0059\,\%$) and $0.0016\,\%$ for 25% (maximum $0.0368\,\%$). The bound held in 180/180 instances; the measured loss was 0.20 to 0.98 of the bound at 25% (median 0.95; the same range at 10%). The bound is nearly attained in six cities and loose by a factor three in Cairo, Moscow and New York; the first inequality of the theorem, which uses the realised aggregate, holds with the measured loss within 10^{-5} of welfare in 180/180 cases. Losses scale with ε^2: a 10% error costs a sixth of a 25% error. With few carriers the market-power charge is itself of the order of welfare, so the same relative errors matter more: scaling it by $1 \pm 25\%$ or $1 \pm 10\%$ leaves the median Nash loss at $2.007\,\%$ and $1.225\,\%$ of the optimum at $N = 6$ (median over both signs; $1.08\,\%$ at $\varepsilon = 0$) and at $0.003\,\%$ and $0.002\,\%$ at $N = 1536$.

11.3. Residents

Residents appear as buyers (consumer surplus) and as people who live near the delivery vehicles. As buyers they are the main gainers from the market-power correction: at $N = 6$ consumer surplus rises by $69.1\,\%$ of welfare under that rule (city medians 41.5% to 86.6 %) because more deliveries are made at a lower price, and it changes by $-1.0\,\%$ under the observation-only charge, which raises the price. At $N = 1536$ both rules take from consumers ($-1.1\,\%$ and $-1.6\,\%$, city range of the observation-only rule -9.8% to $-0.1\,\%$): the charge is passed on in the price, the

congestion cost falls by only −0.1 %, and the council's revenue is the difference. Consumer surplus is 99 % of welfare at $N = 1536$ and 43 % at $N = 6$.

The congestion burden is small in this calibration. Without charge the congestion cost is 1.7 % of welfare at $N = 6$ and 3.6 % at $N = 1536$ (largest city 14.9 %). The delivery vehicles spend on average 0.281 vehicle-minutes per resident and day in the zones they serve at $N = 1536$ (0.160 at $N = 6$, where withholding lowers volume), counting in-zone dwell time only. The charges move this number by −0.6 % (market-power rule) and −0.8 % (observation-only) at $N = 1536$.

Inequality of exposure. Table B.2 uses the GHSL population of each zone. Population only: the repository has no income data, so nothing here concerns income. Exposure is fairly even across the served zones: Gini 0.080 and a top-to-bottom decile ratio of 1.91 at $N = 1536$. At $N = 1536$ the charges leave the Gini unchanged to the third decimal (change +0.000 for the observation-only rule, +0.000 for the market-power rule; 10th to 90th percentile of instances -0.001 to +0.003 for the former). At $N = 6$ the market-power rule lowers the Gini by -0.021 (instances -0.052 to -0.002) and the decile ratio by -0.24; this follows from the +74.1 % larger volume, not from the congestion component of the charge, and the reason the extra volume is spread more evenly was not tested. The observation-only rule changes the Gini by -0.000.

Does the charge push traffic towards dense zones?. In this model it does the opposite. Across the zones of an instance, the relative change of a zone's share of vehicle-hours is negatively rank-correlated with population density (median $\rho_s = -0.56$ for the market-power rule at $N = 1536$, negative in 44/45 instances, 10th to 90th percentile -0.85 to -0.17; −0.59 for the observation-only rule). Dense zones lose share and sparse zones gain it, but the shift is small: the mean log density of exposure moves by -0.002. A plausible reason, checked on three cities (Paris, Cairo, New York, seed 0): the congestion share κ/m of the marginal delivery cost rises with density (rank correlation 0.8 to 0.9), so the charge is relatively larger where demand is less price-sensitive per unit of congestion. The exception is $N = 6$ under the market-power rule, where every zone's volume rises and the correlation is −0.03. The covered zones are the customer zones of the six drawn segments (about 35 zones per instance); the statistics describe those zones, not the whole city.

11.4. Businesses

Businesses here are the delivery carriers; shippers and shops are not modelled. The market-power rule is good for carriers at small N only because of the subsidy: at $N = 6$ carrier profit net of the charge changes by 132.0 % of the no-charge welfare (−181.9 % is the transfer received), while profit before the charge and the delay cost falls by −45.3 % as the lower price outweighs the larger volume, and the delay cost rises by 2.2 %. Under the observation-only rule carriers change by −1.1 % at $N = 6$ (net profit is 57 % of welfare there) and −0.1 % at $N = 1536$. At $N = 1536$ carriers retain only about 1 % of welfare, so this is a loss of roughly 8 % of their profit under the observation-only rule and 7 % under the market-power rule (city medians of the loss, -4.3% to −0.0 % of welfare, observation-only). From the carriers' side the observation-only rule is a small loss at every N; the market-power rule is a large gain at small N, paid for by the council, and a loss of the same size as the observation-only rule once N is large, because the two rules then coincide. Carriers and consumers both gain from the market-power rule at small N, and the council pays.

Sensitivity: congestion ten times higher. With κ multiplied by ten (three seeds, 108 instances; `results/stake_k10.csv`) the congestion cost is 34.0 % of welfare at $N = 1536$ (14.8 % at $N = 6$) and the charges matter more. At $N = 1536$ either rule raises welfare by 2.3 %, collects 24.7 % of the no-charge welfare, cuts congestion cost by −8.8 % and exposure by −11.3 %, and takes −22.1 % from consumers; carrier profit moves by −0.4 %. The Gini of exposure rises by +0.010 and the density correlation stays negative (−0.55). At $N = 6$ the observation-only charge

changes welfare by −3.4 %, and the market-power rule gains 12.3 % at a revenue of −110.8 % of no-charge welfare. The qualitative picture is unchanged: the charge is paid mostly by consumers, and the council's budget is the open question at small N.

Limitations of the stakeholder reading. Consumer surplus is model-based: the area under the linear inverse demand $\alpha_k - b_k Q_k$ calibrated on the delivery fee, not a measured willingness to pay. There are no income data, so the exposure statistics concern population, not who is poor. The number of carriers is fixed in this section; Section 12.2 lets it respond to the charge. The calibrated congestion is weak (congestion cost is 3.6 % of welfare at $N = 1536$), so the congestion side of every table is small and the results mostly describe market power and pass-through. Exposure in this section counts in-zone delivery hours only; Section 12.3 adds driving between zones, emissions and noise. The covered zones are those that the six drawn segments serve.

12. Fixed fees, entry and other externalities

Section 11 left three questions open: who pays for the subsidy that the efficient charge contains when carriers are few, what the subsidy and the charge do to the number of carriers, and what changes when delivery traffic causes costs other than in-zone delay.

12.1. Recovering the subsidy with a fixed fee

Let the city post a charge u per delivery and, in addition, collect a fee T_i from carrier i. Call the fee *lump-sum* if it is fixed before carriers choose their plans: it may depend on the carrier's identity, its licensed fleet or the plan the city is aiming at, but not on the deliveries the carrier actually makes.

Proposition 21 (Lump-sum fees)**.** *Fix the set of N carriers, a charge u and outside options $\underline{\pi}_i$.*

(a) With lump-sum fees the Nash equilibria are those without fees. Deliveries, consumer surplus, congestion cost and welfare are unchanged; the fees move $\sum_i T_i$ from carriers to the city.

(b) All carriers stay in the market iff $\pi_i(u) - T_i \geq \underline{\pi}_i$ for every i, where $\pi_i(u)$ is carrier i's equilibrium profit net of the charge. Fees with $\sum_i T_i = D$ and participation exist iff $\sum_i(\pi_i(u) - \underline{\pi}_i) \geq D$; for the deficit $D = -R$ of the charge and $\underline{\pi}_i = 0$ this reads $\Pi_{\text{op}} - \text{CC} \geq 0$, operating profit net of the delay cost at the target.

(c) A fee that recovers the realised *deficit, $T_i = -\langle u, Q\rangle/N$, is not lump-sum. Carrier i then faces the charge $(1 - \frac{1}{N})u$ on its own deliveries, and in the one-zone case of Proposition 20 the Nash aggregate under $u = u_N^* < 0$ falls below Q^{opt}.*

The proof is in Appendix Appendix A.

The proposition is elementary; for the city, (b) and (c) say that the fee must be computed from the target plan, not from the books after the fact, and must leave each carrier its outside option. Table B.3 applies it to the market-power rule of Section 11 with outside option zero. At $N = 6$ the deficit is 152 % of welfare (median) and the charge is a subsidy in 98 % of zone-hours; recovering it takes 96 % of carrier profit (range 86–101 %). A flat fee per carrier leaves some carrier with a loss in every instance (participation in 0/45), a fee in proportion to the licensed fleet in 4/45 instances; carrier-specific fees exist in 40/45 instances, the exceptions being Buenos Aires, where the subsidy exceeds total carrier profit. At $N = 24$ the fee takes 83 % of profit and a flat fee keeps every carrier in 10/45 instances; at $N = 192$ the charge is a subsidy only in some cities and the fee takes 15 % (participation 38/45 flat); at $N = 1536$ the charge raises revenue and no fee is needed. After the fee, carriers keep 12 % of their no-charge profit at $N = 6$ (28 % at $N = 24$): the subsidy and the fee together transfer the Cournot rent to buyers, and

for $N \leq 192$ no fee scheme leaves every carrier with at least its no-charge profit in any instance. Recovering the realised deficit instead (part (c)) costs 0.10 % of optimal welfare at $N = 6$, small because the residual subsidy is still $(1 - \frac{1}{6})u$.

So the subsidy is budget-balanceable with participation only with carrier-specific fees at $N = 6$, and with simple fees from about $N = 192$ on, where it is small. With a fixed set of carriers this is about distribution; it stops being so when carriers can leave or enter.

12.2. Entry and exit

Model. Segment θ has k_θ identical carriers that share its fleet budget B_θ (Section 9); the fleet is a pool of drivers and vans calibrated from demand, so entry adds firms, not vehicles. Each carrier pays a fixed cost F_θ per day (depot, licence, management; vehicle costs are in the tour cost). Counts are real numbers, a continuous approximation of the integer count; a segment whose single carrier cannot cover F_θ leaves the market. A *free-entry equilibrium* under a charge rule is a Nash equilibrium with $\pi_\theta/k_\theta - T_\theta = F_\theta$ for every active segment, π_θ the segment's profit net of the charge and T_θ its fee per carrier. Welfare is $\mathrm{SW} - \sum_\theta k_\theta F_\theta$.

Proposition 22 (Free entry)**.** *(a) At every free-entry equilibrium, under any charge and lump-sum fees, welfare net of fixed costs equals* $\mathrm{CS} + R + \sum_\theta k_\theta T_\theta$*: consumer surplus plus the city's net revenue. Of two rules that produce the same aggregate, the one with the larger net revenue has the larger welfare.*

(b) One zone, N identical carriers, slack budget, charge u, no fee. Every carrier delivers $s = (F/m)^{1/2}$ whatever u, and the free-entry number is $N^*(u) = (\alpha - u)/(mF)^{1/2} - 1$. Under no charge $N_a = \alpha/(mF)^{1/2} - 1$; under the observation-only rule $N_b = Q_b/s$ with $Q_b = (\alpha - ms)/(m + \kappa)$; under Pigou minus market power ($u = u_N^*$ of Proposition 20 with N endogenous) $N_c = Q^{\mathrm{opt}}/s$; with the fee $T = -uQ^{\mathrm{opt}}/N$ recovering a subsidy, $N_d = \kappa(Q^{\mathrm{opt}})^2/F$, smaller than N_c exactly when the charge is a subsidy ($N_c < m/\kappa$).

(c) (Excess entry.) In (b) with $u = 0$, welfare decreases in N at N_a: a planner who could limit entry, but not change Cournot conduct, would admit fewer carriers.

(d) (Second best, one-zone linear case.) In (b), welfare net of fixed costs as a function of a constant charge u is maximised at $u = \kappa Q_b$, the observation-only fixed point; the market-power term of u_N^* is absent, and u_N^* with endogenous N gives lower welfare.

The proof is in Appendix Appendix A.

Part (c) is the business-stealing result of Mankiw and Whinston (1986); parts (a), (b) and (d) are short computations in the same model. The literature on taxes under free-entry oligopoly finds that the optimal specific tax departs from the Pigou level by a term of either sign that depends on the curvature of demand and costs (Konishi et al., 1990; Katsoulacos and Xepapadeas, 1995; Hamada et al., 2022). Part (d) holds in the one-zone linear case only: output per carrier is pinned down by F alone, a charge changes only the number of carriers, and the departure is zero. The congestion-pricing literature with market power (Brueckner, 2002; Pels and Verhoef, 2004) gives the fixed-N counterpart, a toll below the congestion cost and possibly negative, which is Proposition 20, a finite-N analogue of those results. With N endogenous the market-power subsidy no longer buys output per carrier; it buys entrants.

Calibration of F. F_θ is not observed; we set it so that the no-charge free-entry count lies between k_0 and k_0+1 carriers per segment (the geometric mean of the profit per carrier at k_0 and k_0+1), for $N_0 = 6k_0 \in \{6, 24, 192\}$ (three seeds, 27 city-seed pairs), and sweep $N_0 \in \{12, 48, 96, 384\}$ on seed 0. A smaller N_0 means a larger fixed cost. Zero-profit residuals are below 10^{-5} of F except for the market-power rules, where 119/162 runs reached 10^{-4} and the rest stopped after 80 rounds (residual up to 2e-02 of F, identity of Proposition 22(a) to 2e-02 of welfare); no reported sign depends on them.

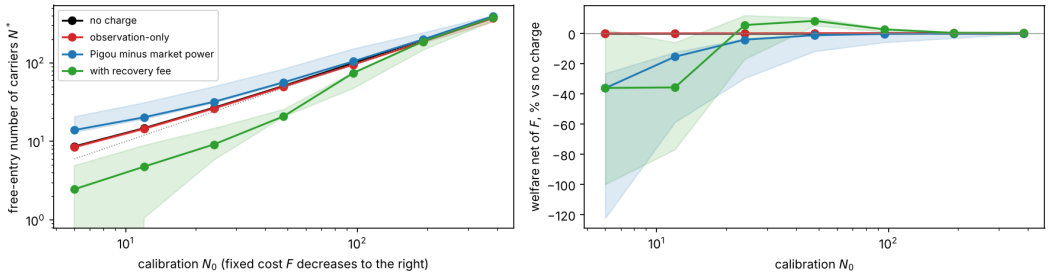


Figure 9: Free entry against the calibration N_0 (medians and interquartile ranges; $N_0 \in \{6, 24, 192\}$ over three seeds, the sweep points on seed 0). Left: free-entry number of carriers. Right: welfare net of fixed costs, % change against no charge.

Results. Table B.4 and Figure 9. *The subsidy attracts entrants.* Under Pigou minus market power the count rises to 14.0 at $N_0 = 6$ and 31.9 at $N_0 = 24$, and welfare net of fixed costs changes by -36.2 % and -4.1 % against no charge (ranges -232.6 to -17.0 % and -74.9 to -2.1 %): the rule that gained 19.6 % with six fixed carriers loses in every instance once entry responds, because the subsidy pays for fixed costs instead of output. *The recovery fee drives exits.* With the fee the count falls to 2.5 at $N_0 = 6$ (median 2 of six segments stay; the market closes in 11/27 instances) and to 9.2 at $N_0 = 24$ (4 segments). The fee raises the cost of being a carrier by about the subsidy each one receives (Proposition 22(b)). Because free entry is excessive (below), the exits can raise welfare: the fee rule beats no charge in 8/27 instances at $N_0 = 6$ and 15/27 at $N_0 = 24$ (median -36.1 and +5.6 %, ranges -100.0 to +23.4 % and -100.0 to +26.2 %); it acts as an uncontrolled entry restriction whose outcome ranges from closing the market to a large gain. *The observation-only charge is safe and close to the second best.* It changes the count little (8.5, 26.4, 186.4), raises welfare and beats the market-power rule in every instance, and is the best of the scalings $\theta \in \{0.5, 1, 1.5\}$ in 27/27, 20/27 and 12/27 instances: the one-zone result of Proposition 22(d) holds approximately, less so with many carriers. The gains are small (+0.0, +0.1 and +0.3 %) because the calibrated congestion is weak. The F sweep in Figure 9 shows the same ordering at every N_0; the fee rule's median changes sign between $N_0 = 12$ and $N_0 = 24$.

Excess entry. Scaling the no-charge free-entry counts by λ (at least one carrier per segment, conduct unchanged), welfare net of fixed costs is largest at one carrier per segment for $N_0 = 6$, at $\lambda = 0.45$ (12 carriers) for $N_0 = 24$ and at $\lambda = 0.24$ (47 carriers) for $N_0 = 192$, gains of 15.87, 11.81 and 3.93 % of welfare; entry is excessive in every instance. Cournot withholding is small once a segment has a few carriers, while each carrier's fixed cost is not. This is the largest welfare effect in this section and it is not a pricing effect: per-delivery charges cannot remove it, and the city's other instrument, a fee on entry, needs the information on profits discussed in Section 12.1.

What is not modelled. Fixed costs are not sunk (entry and exit are free, so N^* is a long-run object) and are common within a segment; with heterogeneous fixed costs inframarginal carriers keep rents and the identity of Proposition 22(a) holds only at the margin. The fleet of a segment is fixed; entrants with their own vehicles would add capacity. The count is continuous; the integer constraint (Mankiw and Whinston, 1986) moves N^* by less than one carrier per segment.

12.3. Emissions, noise and through-traffic

So far the only externality is the delay that delivery vehicles impose on each other while serving a zone; driving between zones is priced only through the tour's own cost. We add emissions, noise and the delay that driving between zones imposes on other traffic as a cost E_j

per tour, linear in its vehicle-km:

$$E_j = \sum_{\text{legs}} \ell_{\text{leg}}\,(e_{\text{air}} + e_{\text{clim}} + e_{\text{noise}}) + \sum_{\text{legs}} t^0_{\text{leg}}\, g(h_{\text{leg}}) + \sum_{\text{stops}} \ell_{\text{stop}}\,(e_{\text{air}} + e_{\text{clim}} + e_{\text{noise}}),$$

with $e.$ in USD per vehicle-km and welfare $\mathrm{SW}_{\text{ext}} = \mathrm{SW} - \sum_i \langle E_i, y_i \rangle$. A leg's length ℓ_{leg} is the length of the fastest path between the two zone nodes on the major-road graph of the processed city data (the same paths as the travel-time matrix of Section 9, reproduced to 10^{-7}), plus an access drive of 120 s at each end. The driving between the drops of one stop, $\ell_{\text{stop}} = 0.7124(A_z\,\text{lot})^{1/2}$, is the Beardwood–Halton–Hammersley length of a tour through lot points spread over the zone's area A_z (Beardwood et al., 1959). Through-traffic delay is the marginal external delay of one vehicle under the BPR curve (Bureau of Public Roads, 1964): a vehicle on a link with free-flow time t^0 at congestion level c adds $\beta c\, t^0$ hours of delay to all others, so $g(h) = \text{pce}\,\beta\,\text{cong}(h)\,\text{VOT}$ per free-flow hour of the leg, at the hourly congestion level of Section 9. It is charged on inter-zone legs only; delay inside a zone is the κ term already in the model. Table B.5 lists the values.

Proposition 23 (Charges for linear tour externalities)**.** *Let tour j of carrier i cause the external cost E_{ij}, and let Q' be the aggregate of an extended social optimum (maximiser of SW_{ext}).*

(a) A charge E_{ij} per tour (a distance charge) plus $\kappa \odot Q'$ per delivery makes the mean-field equilibrium an extended optimum; for N identical carriers, a charge E per tour plus $(\kappa - \frac{m}{N}) \odot Q'$ per delivery makes the Nash equilibrium one.

(b) If the external cost is additive over deliveries, $E_i = L_i^\top e$ for a common $e \in \mathbb{R}^d$, the per-delivery charge $\kappa \odot Q' + e$ alone does the same in the mean field.

The proof is in Appendix Appendix A.

Vehicle-km are not proportional to deliveries, so (b) holds only approximately. We compute, in the mean field, the exact rule of (a); the *matched* per-delivery charge that reproduces Q' (damped iteration as in Section 11); and the simple rule $u = \kappa \odot Q' + \bar{e}$, with $\bar{e}_k$ the external cost of the tours delivering at k per delivery made there; at finite N, the market-power rule with these terms.

Results. Table B.6 and Figure 10. A delivery drives 1.53 vehicle-km (median; 31 % of it inside the delivery zones) and causes 0.405 USD of external cost beyond in-zone delay (city medians 0.054–1.552), 15 % of the delivery fee (5–37 %). Through-traffic delay is 84 % of it; emissions are 0.011 (air) and 0.039 (climate) USD per delivery and noise 0.008. The external cost is 8.0 times the in-zone congestion charge κQ (city medians 3.7–36.6). This is the main finding of the extension and a negative one for the model of the earlier sections: the congestion term that the model prices, the delay among delivery vehicles while they serve a zone, is the smaller part of the congestion they cause.

The charge. The matched per-delivery charge is 0.50 USD per delivery against 0.048 for the congestion-only Pigou charge (median ratio 8, by city 5–33). At the extended optimum deliveries fall by -17 % and vehicle-km by -29 % (-13 % per delivery: carriers switch to tours with fewer km per drop). *Welfare.* Without a charge the market loses 5.43 % of the optimal welfare of Section 11 to the extended optimum (above 20 % in New Delhi, Mumbai and London); the congestion-only Pigou charge recovers 5 % of this gap (at most 17 %), the matched per-delivery charge all but 0.08 % (at most 0.8 %), and the simple rule $\kappa \odot Q' + \bar{e}$ all but 1.43 %: averaging the cost of a tour over its deliveries does not reproduce the optimal aggregate (deliveries +3 % and vehicle-km +12 % against the extended optimum). Zone-hour charges therefore suffice for these externalities if they are set to match the right aggregate; a distance charge adds little. With few carriers market power still dominates: at $N = 6$ the losses are 11.11 % (no charge), 6.91 % (market-power rule, congestion only), 2.22 % (extended, per delivery) and 1.40 % (extended, with distance charge); at $N = 1536$ they are 5.36, 5.14, 0.10 and 0.00 %.

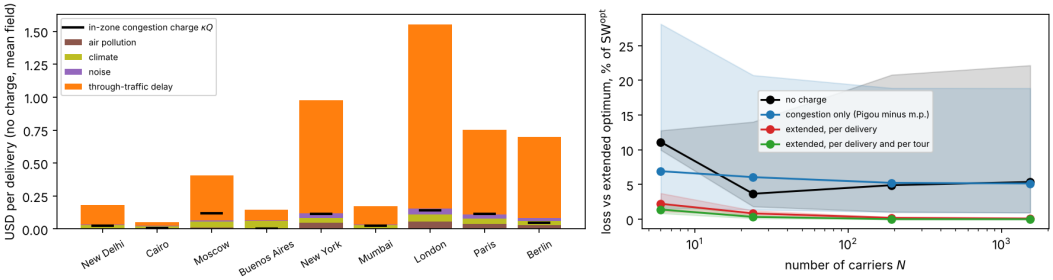


Figure 10: Left: external cost per delivery by component and city (no charge, mean field, median of five seeds), with the in-zone congestion charge κQ (black). Right: welfare below the extended optimum against N (% of the optimal welfare of Section 11; median and interquartile range over 45 instances).

Sensitivity. European values everywhere: 0.461 USD per delivery, no-charge loss 5.39 %; Euro 4 air pollution and the EPA carbon value: 0.440 and 5.84 %; emissions and noise only: 0.066 and 0.16 %. Emissions and noise alone change the earlier conclusions very little; the through-traffic term carries the result, and its value of time (the driver wage applied to all traffic) is the weakest parameter of Table B.5.

Exposure. Assigning each leg's km to the zones its path crosses and each stop's km to its zone, delivery traffic puts 46 vehicle-metres per resident and day on the zones it touches (city medians 7–100; 32 % in zones that receive no deliveries), an air-pollution damage of 0.26 USD per resident and year (at most 1.45). This exposure is far less even than the in-zone exposure of Section 11.3: Gini 0.47 against 0.080, and a top-to-bottom decile ratio of 93, consistent with through traffic running on a few corridors. Across served zones it is positively but loosely rank-correlated with the in-zone exposure (Spearman 0.49, city medians 0.36–0.65). The extended charge cuts this exposure by -29 % (congestion-only Pigou: -0.7 %) but raises its Gini to 0.50 (higher in 42/45 instances); why the remaining traffic is more concentrated was not tested. Population is the only weight: there are no income or health data, and pollution damage is per vehicle-km, not modelled from concentrations.

13. Discussion

What is proved, what is numerically supported, what is open. Proved, with complete proofs here (in the text or in the appendix): every lemma, theorem, proposition and corollary stated, except Conjecture 19. Theorem 8 rests on Robinson (1981), Proposition 15(a) and Corollary 16 on the companion paper, and Proposition 11(a) restates the contraction theorem of the companion paper. Every bound was also checked by brute force on small random games (Appendix C). Numerically supported only: that the city instances behave like $O(s_N)$ with a constant near one, and that heterogeneous splits do too; that with several carrier types the observation-only charge stays close to the second best under free entry (Proposition 22(d) is for one zone with linear costs); and all magnitudes of Section 12. Open: an explicit constant for replicas in the polyhedral case, and Conjecture 19.

Who gains and who loses. Section 11 splits welfare into carrier profit, consumer surplus, congestion cost and charge revenue. With few carriers the efficient charge is a subsidy that residents (as buyers) and carriers welcome and the council pays for; with many carriers all rules converge to the Pigou charge, which raises a few per cent of welfare and falls on consumers and, in proportion, on carrier profit. The congestion burden for residents is small in the calibration, so the distributional findings are mostly about market power and pass-through. Funding the subsidy from carriers is possible at six carriers only with carrier-specific fees, and once carriers can enter and leave the subsidy buys entrants and the fee drives exits (Section 12).

What the results mean for a city. With many small carriers, a city can set congestion charges from observed zone-hour totals and approach the marginal-cost charge; welfare converges at rate $O(\bar{s})$ regardless of how carriers' tours couple zones. With few carriers this rule is harmful, and the market-power correction of the companion paper is needed. Cautious carriers make an observation-only city over-charge by about their caution margin; the welfare cost is second order. If carriers can enter freely, the market-power subsidy was counterproductive in the calibrated cities, and the plain observation-only charge was the better rule there; the second-best property of that charge is proved for one zone with linear costs only. Most of the external cost of delivery traffic is the delay it causes to other traffic between zones, not the in-zone delay the base model prices; zone-hour charges set to the extended marginal cost and matched to the right aggregate capture nearly all of it.

Limitations. The hourly profile is stylised; real hourly demand and speed data would change the levels in E2–E4 and possibly the economic conclusions, but not the convergence statements. Travel times are deterministic in the experiments (the p-box routing layer of the companion paper is not re-run on the time-expanded graph, and Proposition 17 shows that its corner property can fail with time-dependent charges). In-zone congestion is linear in in-zone hours; driving between zones enters only as a linear external cost on other traffic (Section 12.3), valued with a proxy value of time, and emissions and noise use unit values per vehicle-km transferred from European estimates. Fixed costs are not sunk and are common within a segment. Tours have at most three stops. The segments are drawn, not observed. The mean-field limit removes market power, which is the economic content of the leader results of the companion paper; for markets with a few large carriers and many small ones, a major-minor model is the right object and is left for later work.

14. Conclusion

The mean-field limit of competitive urban delivery is reached at rate $O(\bar{s})$ when carriers' problems separate across zone-hours, and in the city instances; in the worst case, once tours and fleets couple zones and strong monotonicity in individual strategies fails, only the square-root rate holds, and an explicit example attains it; under strong monotonicity the $O(1/M)$ bounds of Paccagnan et al. (2019) apply. The city's observation-only charge inherits these rates, but its efficiency converges at rate $O(\bar{s})$ for every instance. Whether a constant that depends only on the number of zone-hours exists for natural subclasses (unweighted fleet budgets, for example) is the main open question.

Appendix A. Proofs for Section 12

Proof of Proposition 21. (a) Carrier i's payoff $J_i(y_i; y_{-i}) - T_i$ differs from J_i by a constant in y_i, so best responses coincide. Welfare (2) depends on (π_i) and Q only. (b) The first claim is the definition. If $\sum_i(\pi_i - \underline{\pi}_i) \geq D$, the fees $T_i = \pi_i - \underline{\pi}_i - \frac{1}{N}(\sum_j(\pi_j - \underline{\pi}_j) - D)$ sum to D and leave every carrier at least $\underline{\pi}_i$; the converse follows by summing the participation constraints. With $D = -R$ and $\Pi = \Pi_{\mathrm{op}} - \mathrm{CC} - R$ (Section 11) the condition is $\Pi_{\mathrm{op}} - \mathrm{CC} \geq \sum_i \underline{\pi}_i$. (c) $T_i = -\frac{1}{N}\langle u, A_{-i}\rangle - \frac{1}{N}\langle u, a_i\rangle$, so $J_i - T_i$ equals $J_i + \frac{1}{N}\langle u, a_i\rangle$ up to a constant in y_i. In one zone the Nash condition $\alpha - (1 - \frac{1}{N})u = (1 + \frac{1}{N})mQ$ gives $Q < Q^{\mathrm{opt}}$ when $u < 0$. □

Proof of Proposition 22. (a) $\mathrm{SW} = \mathrm{CS} + \Pi + R$ (Section 11) with Π net of the charge. With fees, carriers keep $\Pi - \sum_\theta k_\theta T_\theta$ and zero profit gives $\Pi - \sum_\theta k_\theta T_\theta = \sum_\theta k_\theta F_\theta$; inactive segments contribute nothing. $\mathrm{CS} = \frac{1}{2}\langle b, Q^2\rangle$ depends on Q only. (b) The Nash condition is $\alpha - u - mQ = mq$ and the profit of a carrier is $(\alpha - u - mQ)q = mq^2$ (Proposition 20); zero profit gives $q = s$, and $Q = Nq$ gives N^*. For $u = \kappa Q$: $Q(m + \kappa) = \alpha - ms$. Under u_N^* the aggregate is Q^{opt} and $q = s$. With the fee, profit is $mq^2 + uq = (\alpha - mQ^{\mathrm{opt}})q = \kappa Q^{\mathrm{opt}} q$, so $q = F/(\kappa Q^{\mathrm{opt}})$ and

$N_d = \kappa(Q^{\mathrm{opt}})^2/F$; $N_d/N_c = \kappa Q^{\mathrm{opt}}/(ms) = \kappa N_c/m$. (c) With $Q(N) = \frac{N}{N+1}\frac{\alpha}{m}$ and $q = Q/N$, $Q'(N) = q/(N+1)$ and $\frac{d}{dN}(\mathrm{SW} - NF) = (\alpha - (m+\kappa)Q)Q' - F = (mq - \kappa Q)\frac{q}{N+1} - mq^2 < 0$ at $F = mq^2$. (d) Since $q = s$ for every u, $N = Q/s$ and $\mathrm{SW}(Q) = \alpha Q - \frac{b}{2}Q^2 - \kappa Q^2 - msQ$, concave with maximiser Q_b; the map $u \mapsto Q = (\alpha - u)/m - s$ is a bijection and $Q = Q_b$ at $u = \kappa Q_b$. □

Proof of Proposition 23. (a) $\mathrm{SW}_{\mathrm{ext}}$ is the welfare of the game with tour profits $p_i - E_i$, which satisfies (A1)–(A2); a carrier that pays E per tour has exactly these profits. The mean-field statement is the Pigou property of that game (Lemma 2 with $u = \kappa \odot Q'$) and the N-carrier statement is Proposition 20 applied to it. (b) $\langle E_i, y_i\rangle = \langle e, L_i y_i\rangle = \langle e, a_i\rangle$ for every y_i, so a charge e per delivery is the distance charge of (a). □

Appendix B. Secondary tables

Parameters and per-city or per-size tables that support Sections 9, 11 and 12.

Table B.1: Replica study per city (amplitude 1, five seeds). d: active zone-hours; $\varrho = \max \kappa/m$; slope: log-log slope of $\|Q^{\mathrm{N}} - Q^{\mathrm{W}}\|_m$ in N for $N \geq 24$, mean over seeds with 95% t-interval; ratio: $\max_k |\Delta Q_k|/s_N$ with s_N the largest Nash delivery of one carrier at one zone-hour, median and maximum over N and seeds; PoA−1 in % of optimal welfare at $N = 6$ and in the limit; charge gap $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}}/\|u^*_{\mathrm{W}}\|_{\kappa^{-1}}$ at $N = 6$ and $N = 1536$.

City	d	ϱ	slope [95% CI]	ratio med	ratio max	PoA_6	PoA_∞	charge_6	charge_{1536}
New Delhi	290	0.209	-0.92 [-0.95, -0.90]	1.02	1.53	22.4	0.357	0.424	0.0036
Cairo	179	0.009	-0.77 [-0.81, -0.73]	0.97	1.65	9.6	0.001	0.266	0.0025
Moscow	312	0.074	-0.93 [-0.96, -0.90]	0.73	1.55	23.3	0.026	0.414	0.0023
Buenos Aires	321	0.019	-0.92 [-0.94, -0.89]	0.99	1.62	26.9	0.003	0.453	0.0039
New York	295	0.025	-0.78 [-0.87, -0.69]	0.72	1.86	18.4	0.004	0.337	0.0027
Mumbai	247	0.154	-0.90 [-0.92, -0.89]	1.14	1.62	19.8	0.224	0.403	0.0038
London	316	0.083	-0.92 [-0.95, -0.89]	0.98	1.23	23.1	0.201	0.440	0.0038
Paris	322	0.042	-0.86 [-0.90, -0.83]	1.06	1.68	17.7	0.037	0.363	0.0037
Berlin	346	0.033	-0.92 [-0.97, -0.87]	0.94	1.82	25.0	0.010	0.443	0.0033

Table B.2: Exposure of residents of the served zones (median of 45 instances). Exposure: delivery vehicle-hours in the resident's zone per resident, scaled to the real city (in-zone dwell only). Mean: vehicle-minutes per resident and day, population-weighted; Gini: population-weighted Gini of daily exposure by zone, and of exposure by zone-hour; D10/D1: mean exposure of the most exposed tenth of residents over that of the least exposed tenth; ρ_s: Spearman correlation across zones, within an instance, between the relative change of a zone's share of vehicle-hours (from no charge) and the log of its population density.

N	rule	mean	Gini zone	Gini zone-hour	D10/D1	ρ_s
6	no charge	0.160	0.135	0.314	2.25	–
	observation-only	0.159	0.134	0.313	2.22	-0.56
	Pigou minus market power	0.278	0.080	0.289	1.90	-0.03
1536	no charge	0.281	0.080	0.291	1.91	–
	observation-only	0.277	0.080	0.288	1.91	-0.59
	Pigou minus market power	0.278	0.080	0.288	1.91	-0.56

Table B.3: Budget-balanced market-power rule at a fixed number of carriers (five seeds per city). For each N: fee as % of carrier profit before the fee (median), and the number of seeds with participation (every carrier's profit after the fee ≥ 0) under a flat fee / a fee in proportion to the licensed fleet / carrier-specific fees (the aggregate condition of Proposition 21(b)). "tax": the median charge raises revenue, no fee needed.

	$N = 6$		$N = 24$		$N = 192$		$N = 1536$	
city	fee %	part.	fee %	part.	fee %	part.	fee %	part.
New Delhi	95	0/0/5	76	0/0/5	tax		tax	
Cairo	94	0/2/5	85	3/5/5	58	4/5/5	4	5/5/5
Moscow	94	0/0/5	80	4/4/5	5	5/5/5	tax	
Buenos Aires	101	0/0/0	99	0/0/5	81	0/0/5	tax	
New York	88	0/2/5	69	3/5/5	18	5/5/5	tax	
Mumbai	97	0/0/5	79	0/0/5	tax		tax	
London	96	0/0/5	81	0/1/5	tax		tax	
Paris	98	0/0/5	90	0/0/5	18	5/5/5	tax	
Berlin	98	0/0/5	91	0/0/5	36	4/4/5	tax	

Table B.4: Free entry (nine cities, three seeds; medians). N^*: free-entry number of carriers (real); seg.: active segments of six; ΔSW: welfare net of fixed costs, % change against no charge (free entry in both); R: city's net revenue (charge plus fee), % of no-charge welfare. $\theta\kappa Q$: observation-only rule with the charge scaled by θ.

	$N_0 = 6$				$N_0 = 24$				$N_0 = 192$			
rule	N^*	seg.	ΔSW	R	N^*	seg.	ΔSW	R	N^*	seg.	ΔSW	R
no charge	8.6	6	0	+0.0	26.9	6	0	+0.0	195.0	6	0	+0.0
observation-only	8.5	6	+0.0	+3.5	26.4	6	+0.1	+3.5	186.4	6	+0.3	+3.5
Pigou minus market power	14.0	6	-36.2	-165.7	31.9	6	-4.1	-34.6	198.9	6	-0.1	-2.2
with recovery fee	2.5	2	-36.1	+0.0	9.2	4	+5.6	-0.0	187.2	6	+0.3	+0.0
$0.5\,\kappa Q$	8.6	6	+0.0	+1.8	26.6	6	+0.0	+1.8	190.6	6	+0.2	+1.8
$1.5\,\kappa Q$	8.4	6	+0.0	+5.2	26.2	6	+0.1	+5.1	182.5	6	+0.2	+5.1

Table B.5: External cost parameters, USD 2025 per vehicle-km (EUR 2016 values of van Essen et al. (2020) times 1.285, euro-area HICP 2016–2025, and 1.122 USD per EUR). Grades as in Section 9. Air pollution and noise are transferred to the non-European cities by the driver wage relative to Paris, Berlin and London (income elasticity one, grade C); air pollution and climate are scaled by $\frac{1}{3}$ for the small vehicles of New Delhi, Mumbai and Cairo (the model's fuel ratio, C).

item	value	source	grade
air pollution	0.032	diesel Euro 6 LCV, metropolitan urban road, 2.19 EURct (van Essen et al., 2020, Tab. 19); sensitivity Euro 4	B
climate	0.035	same vehicle, 2.40 EURct at EUR 100/t CO_2e (van Essen et al., 2020, Tab. 28); sensitivity USD 190/t of 2020 (U.S. Environmental Protection Agency, 2023)	B
noise	0.025	LCV, day, dense urban traffic, 1.7 EURct (van Essen et al., 2020, Tab. 37)	B
through-traffic delay	pce β cong(h) VOT per free-flow hour	BPR $\beta = 4$, pce 1.5, hourly congestion of Section 9; VOT of other traffic = driver wage per vehicle-hour. Cross-check: LCVs on urban roads near to over capacity, 0.26–0.99 EUR/vkm (van Essen et al., 2020, Tab. 118); ours 0.76–1.14 USD per inter-zone vkm in London, Paris, Berlin	C
access drive	0.67 km per trip end	120 s (access time of the travel-time matrix) at 20 km/h	C
in-zone driving	$0.7124\sqrt{A_z \text{ lot}}$	tour length through uniformly spread drops (Beardwood et al., 1959)	B

Table B.6: External costs per city (mean field, no charge unless stated; median of five seeds). Emiss.: air pollution and climate; noise; through: through-traffic delay; κQ: the in-zone congestion charge, all in USD per delivery; %fee: total external cost per delivery as % of the delivery fee. Loss: welfare below the extended optimum, % of the optimal welfare of Section 11, with no charge, with the congestion-only Pigou charge and with the matched per-delivery charge. Δvkm: change of vehicle-km at the extended optimum against no charge. Exposure: vehicle-metres per resident and day in the zones the traffic touches (no charge), and its population-weighted Gini without charge and at the extended optimum.

	USD per delivery					loss, % of SW^{opt}				exposure		
city	emiss.	noise	through	κQ	%fee	none	Pigou	matched	Δvkm %	m/res.	Gini	Gini′
New Delhi	0.029	0.004	0.151	0.027	27	26.56	21.99	0.03	-53	7	0.44	0.53
Cairo	0.021	0.001	0.031	0.007	5	0.12	0.14	0.07	-3	11	0.27	0.28
Moscow	0.057	0.008	0.339	0.117	11	1.01	1.04	0.10	-12	87	0.48	0.49
Buenos Aires	0.061	0.003	0.084	0.004	15	6.80	6.59	0.06	-31	7	0.51	0.57
New York	0.085	0.035	0.857	0.113	11	0.47	0.52	0.15	-8	100	0.51	0.51
Mumbai	0.029	0.004	0.139	0.023	25	22.33	18.84	0.05	-49	9	0.35	0.42
London	0.112	0.043	1.397	0.140	37	35.16	31.46	0.06	-57	49	0.55	0.62
Paris	0.081	0.028	0.644	0.113	12	4.57	4.22	0.15	-29	49	0.42	0.47
Berlin	0.061	0.023	0.615	0.049	16	5.12	4.84	0.07	-27	62	0.49	0.53

Appendix C. Numerical checks of the theory

The script `mfg/checks/theory_checks.py` computes, on small random games with linear profits and fleet budgets (and by exhaustive enumeration for pure equilibria of lumpy games), the slack of every inequality of Sections 3–7; its output is `theory_checks.out`. All slacks were nonnegative (up to the solver accuracy). In particular: the linearisation of Lemma 2(c) reproduces the Nash aggregate to 10^{-7}; the bound of Theorem 4(a) held in 150 games with largest ratio 0.92; the closed form of Proposition 5 matches the solver; Theorem 6 held in 200 convexified and 300 lumpy one-zone games (358 pure equilibria), with the constant one attained; Theorem 4(b) held on 287 pure equilibria of multi-zone lumpy games.

Appendix D. Machine-checked cores

The Lean 4 file `mfg/lean/MFG/Core.lean` (Mathlib; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound` only) checks: the symmetric closed form of Theorem 6; the inner-product algebra of Theorem 4(a) from the per-carrier optimality inequalities; the revealed-preference and sandwich steps of Theorem 6; the contraction perturbation lemma of Theorem 12; the weighted-norm chain of Proposition 11(a); the algebra of the tightness family (Proposition 5); the welfare inequalities behind Theorem 13(b) and Proposition 15(d); the determinant and the weight equivalence of Proposition 18; and the width bound of Corollary 16. It does not formalise existence, first-order optimality, Robinson's theorem or the theorems of the companion paper that are used.

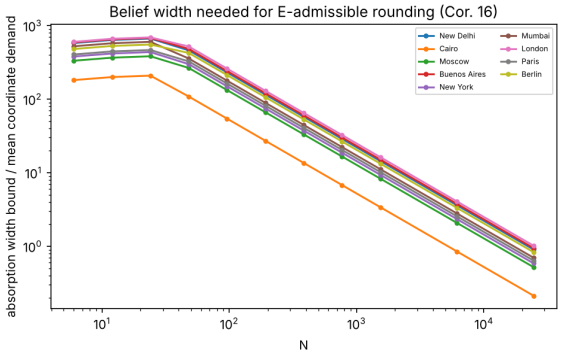


Figure D.11: Uniform belief width of Corollary 16 relative to the mean demand of a zone-hour (seed 0).

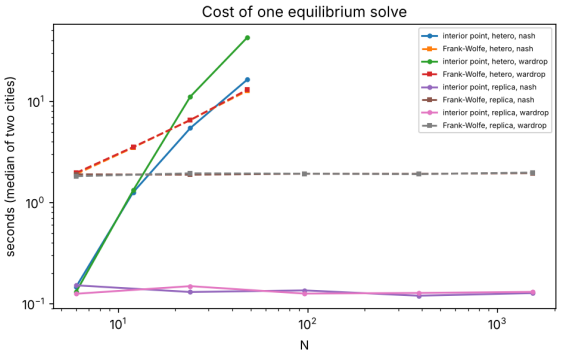


Figure D.12: Time of one equilibrium solve (interior point and Frank–Wolfe) for replicas and heterogeneous splits.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Code, instances, result files and the Lean development are available at https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty (folder `mfg`); the city data are the processed data of the companion paper. The repository is not a persistent archive with a DOI.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to support the writing of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

Al Saleh, F., Bakaryan, T., Gomes, D.A., de Lima Ribeiro, R., 2024. First-order mean-field games on networks and Wardrop equilibrium. Portugaliae Mathematica 81, 201–246.

Arslan, A.M., Agatz, N., Kroon, L., Zuidwijk, R., 2019. Crowdsourced delivery—a dynamic pickup and delivery problem with ad hoc drivers. Transportation Science 53, 222–235.

Beardwood, J., Halton, J.H., Hammersley, J.M., 1959. The shortest path through many points. Mathematical Proceedings of the Cambridge Philosophical Society 55, 299–327.

Beckmann, M., McGuire, C.B., Winsten, C.B., 1956. Studies in the Economics of Transportation. Yale University Press.

Brueckner, J.K., 2002. Airport congestion when carriers have market power. American Economic Review 92, 1357–1375.

Buchanan, J.M., 1969. External diseconomies, corrective taxes, and market structure. American Economic Review 59, 174–177.

Bureau of Public Roads, 1964. Traffic Assignment Manual for Application with a Large, High Speed Computer. Technical Report. U.S. Department of Commerce, Urban Planning Division. Washington, DC.

Cabannes, T., Laurière, M., Pérolat, J., et al., 2021. Solving N-player dynamic routing games with congestion: a mean field approach. arXiv preprint arXiv:2110.11943.

Calderone, D., Sastry, S.S., 2017. Markov decision process routing games, in: Proceedings of the 8th International Conference on Cyber-Physical Systems (ICCPS), pp. 273–279.

Cardaliaguet, P., Delarue, F., Lasry, J.M., Lions, P.L., 2019. The Master Equation and the Convergence Problem in Mean Field Games. volume 201 of *Annals of Mathematics Studies*. Princeton University Press.

Cardaliaguet, P., Hadikhanloo, S., 2017. Learning in mean field games: the fictitious play. ESAIM: Control, Optimisation and Calculus of Variations 23, 569–591.

Carmona, R., Delarue, F., 2018. Probabilistic Theory of Mean Field Games with Applications I–II. Probability Theory and Stochastic Modelling, Springer.

Cattaruzza, D., Absi, N., Feillet, D., González-Feliu, J., 2017. Vehicle routing problems for city logistics. EURO Journal on Transportation and Logistics 6, 51–79.

Cominetti, R., Scarsini, M., Schröder, M., Stier-Moses, N., 2023. Approximation and convergence of large atomic congestion games. Mathematics of Operations Research 48, 784–811.

Dang, Q.V., 2026. City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search. Manuscript under review.

Doncel, J., Gast, N., Gaujal, B., 2019. Discrete mean field games: existence of equilibria and convergence. Journal of Dynamics and Games 6, 221–239.

Eliasson, J., Mattsson, L.G., 2006. Equity effects of congestion pricing: quantitative methodology and a case study for Stockholm. Transportation Research Part A: Policy and Practice 40, 602–620.

van Essen, H., van Wijngaarden, L., Schroten, A., Sutter, D., Bieler, C., Maffii, S., Brambilla, M., Fiorello, D., Fermi, F., Parolin, R., El Beyrouty, K., 2020. Handbook on the External Costs of Transport, Version 2019 – 1.1. Technical Report. European Commission, Directorate-General for Mobility and Transport; Publications Office of the European Union. Luxembourg. Doi:10.2832/51388.

Gilboa, I., Schmeidler, D., 1989. Maxmin expected utility with non-unique prior. Journal of Mathematical Economics 18, 141–153.

Gomes, D.A., Mohr, J., Souza, R.R., 2010. Discrete time, finite state space mean field games. Journal de Mathématiques Pures et Appliquées 93, 308–328.
Gomes, D.A., Mohr, J., Souza, R.R., 2013. Continuous time finite state mean field games. Applied Mathematics & Optimization 68, 99–143.
Goulart, P.J., Chen, Y., 2024. Clarabel: an interior-point solver for conic programs with quadratic objectives. arXiv preprint arXiv:2405.12762 .
Guéant, O., 2015. Existence and uniqueness result for mean field games with congestion effect on graphs. Applied Mathematics & Optimization 72, 291–303.
Guo, X., Hu, A., Zhang, J., 2022. Optimization frameworks and sensitivity analysis of Stackelberg mean-field games. arXiv preprint arXiv:2210.04110.
Hall, J.V., Krueger, A.B., 2018. An analysis of the labor market for Uber's driver-partners in the United States. ILR Review 71, 705–732.
Hamada, K., Ohkawa, T., Okamura, M., 2022. Optimal taxation in a free-entry Cournot oligopoly: the average cost function approach. Canadian Journal of Economics 55, 1166–1192.
Haurie, A., Marcotte, P., 1985. On the relationship between Nash-Cournot and Wardrop equilibria. Networks 15, 295–308.
Huang, M., 2010. Large-population LQG games involving a major player: the Nash certainty equivalence principle. SIAM Journal on Control and Optimization 48, 3318–3353.
Huang, M., Malhamé, R.P., Caines, P.E., 2006. Large population stochastic dynamic games: closed-loop McKean-Vlasov systems and the Nash certainty equivalence principle. Communications in Information and Systems 6, 221–252.
Iyengar, G.N., 2005. Robust dynamic programming. Mathematics of Operations Research 30, 257–280.
Jacquot, P., Wan, C., 2018. Routing game on parallel networks: the convergence of atomic to nonatomic. arXiv preprint arXiv:1804.03081.
Katsoulacos, Y., Xepapadeas, A., 1995. Environmental policy under oligopoly with endogenous market structure. Scandinavian Journal of Economics 97, 411–420.
Konishi, H., Okuno-Fujiwara, M., Suzumura, K., 1990. Oligopolistic competition and economic welfare: a general equilibrium analysis of entry regulation and tax-subsidy schemes. Journal of Public Economics 42, 67–88.
Langner, J., Neufeld, A., Park, K., 2024. Markov-Nash equilibria in mean-field games under model uncertainty. arXiv:2410.11652.
Lasry, J.M., Lions, P.L., 2007. Mean field games. Japanese Journal of Mathematics 2, 229–260.
Lavigne, P., Pfeiffer, L., 2023. Generalized conditional gradient and learning in potential mean field games. Applied Mathematics & Optimization .
Leape, J., 2006. The London congestion charge. Journal of Economic Perspectives 20, 157–176.
Li, Y., Dimakis, A., Courcoubetis, C.A., 2025. Repositioning, ride-matching, and abandonment in on-demand ride-hailing platforms: a mean field game approach. arXiv preprint arXiv:2504.02346.
Mankiw, N.G., Whinston, M.D., 1986. Free entry and social inefficiency. RAND Journal of Economics 17, 48–58.
Monderer, D., Shapley, L.S., 1996. Potential games. Games and Economic Behavior 14, 124–143.
Novshek, W., 1985. Perfectly competitive markets as the limits of Cournot markets. Journal of Economic Theory 35, 72–82.
Paccagnan, D., Gentile, B., Parise, F., Kamgarpour, M., Lygeros, J., 2019. Nash and Wardrop equilibria in aggregative games with coupling constraints. IEEE Transactions on Automatic Control 64, 1373–1388.
Pels, E., Verhoef, E.T., 2004. The economics of airport congestion pricing. Journal of Urban Economics 55, 257–277.
Pigou, A.C., 1920. The Economics of Welfare. Macmillan, London.
Robinson, S.M., 1981. Some continuity properties of polyhedral multifunctions. Mathematical Programming Study 14, 206–214.
Sanjari, S., Bose, S., Başar, T., 2022. Incentive designs for Stackelberg games with a large number of followers and their mean-field limits. arXiv preprint arXiv:2207.10611.
Slade, M.E., 1994. What does an oligopoly maximize? The Journal of Industrial Economics 42, 45–61.
U.S. Environmental Protection Agency, 2023. Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances. Technical Report. U.S. Environmental Protection Agency. Washington, DC.
Verhoef, E.T., 2002. Second-best congestion pricing in general networks. Heuristic algorithms for finding

second-best optimal toll levels and toll points. Transportation Research Part B: Methodological 36, 707–729.

Wardrop, J.G., 1952. Some theoretical aspects of road traffic research. Proceedings of the Institution of Civil Engineers, Part II 1, 325–362.

Yang, H., Huang, H.J., 2005. Mathematical and Economic Theory of Road Pricing. Elsevier, Oxford.

Yang, H., Meng, Q., Lee, D.H., 2004. Trial-and-error implementation of marginal-cost pricing on networks in the absence of demand functions. Transportation Research Part B: Methodological 38, 477–493.